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A remote sensing-based approach to suspended sediment estimation in rivers with scarce in situ data

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ABSTRACT

Monitoring suspended sediment concentration (SSC) is essential for water-quality management and for anticipating sediment-related impacts. Yet sparse in situ measurements limit tracking SSC at high temporal and spatial resolution. We address this gap with a scalable approach that couples multispectral imagery (Landsat-8/9, Sentinel-2), remote sensing precipitation product (CHIRPS), and soil properties product (SoilGrids) within artificial neural networks (ANNs) to estimate SSC in the Doce and Upper São Francisco River basins, Brazil. Data acquisition and preprocessing were performed in Google Earth Engine, and ANNs were trained in MATLAB and ported to GEE for pixel-wise mapping. We compiled 978 SSC observations from 41 gauging stations (rivers >30 m wide) to train and validate three model families: reflectance-only, precipitation/soil-only, and an integrated model. Reflectance-based ANNs best captured peaks and event-driven anomalies, including non-rainfall disturbances, while the precipitation/soil model produced stable daily estimates during cloudy periods, reproducing seasonal cycles. The integrated model reconciled both controls and achieved the best overall skill (R^2 up to 0.84; MAE ≈ 18.7 mg L^{-1}). The results show that precipitation, soil properties, and reflectance are complementary predictors that can be combined to deliver continuous, spatially explicit SSC information in data-scarce basins. The workflow is cost-effective, reproducible, and readily deployable for routine monitoring, post-event assessment, and planning of erosion control and reservoir siltation management.

1. Introduction

Monitoring suspended sediment concentration (SSC) in rivers is essential for evaluating processes such as erosion, sediment transport and deposition, reservoir siltation, and water quality in river basins (Vercruysse et al., 2017). SSC responds to land use changes, soil erosion alterations, and hydraulic infrastructure construction, and is also sensitive to intense rainfall events that can increase the transport of particles into water bodies. By linking watershed drivers to sedimentation, SSC monitoring reveals the distribution and variability of sediments, which is crucial for tackling water quality issues and guiding water resource management in rivers and estuaries (Dethier et al., 2022). Despite its importance, monitoring SSC in practice remains challenging. Traditionally, SSC monitoring is performed through in situ sampling, a method that requires human and financial resources, along with laboratory analyses, making it an expensive process. As a result, only a few rivers within the same region are monitored, and even in these rivers the temporal resolution is extremely limited, often consisting of only two to four SSC measurements per year. This scarcity of data constrains the characterization of the spatial distribution of SSC as well as its seasonal and interannual variability (Campos and Pedrollo, 2021). In this context, indirect approaches such as remote sensing become a valuable alternative for expanding SSC information in both space and time within data-limited regions.

Over the past decades, one of the most widely used and straightforward indirect methods for estimating SSC in rivers has been the

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application of SSC rating-curves (Asselman, 2000; Heng and Suetsugi, 2013; Lohani et al., 2007; Zhang et al., 2012). These empirical regressions are typically derived for individual gauging stations from observed relationships between discharge and SSC. However, the SSC–discharge relationship is often non-linear, and hysteresis effects commonly reduce the accuracy of this method. In addition to these empirical approaches, more complex conceptual and physically based models have also been employed to estimate SSC, incorporating hydrological and sediment transport processes more explicitly (Al Khoury et al., 2023; Aloui et al., 2023; Fagundes et al., 2023, 2021, 2020). The main obstacle for the conceptual and physical models is that they require high-resolution and high-quality information on flow and sediment, which is unavailable in most cases (Nourani et al., 2012).

Machine-learning (ML) models are a growing alternative for SSC estimation (Aires et al., 2023; Buyukyildiz and Kumcu, 2017; Campos and Pedrollo, 2023; Choubin et al., 2018; Heng and Suetsugi, 2013; Nguyen et al., 2024), as they can capture non-linear and complex relationships between variables without relying on explicit physical equations. Their ability to learn complex relationships directly from data makes ML models particularly suitable for SSC estimation, since sediment dynamics result from interacting hydrological, geomorphological, and meteorological drivers that vary in space and time. Within the broad family of ML techniques, artificial neural networks (ANNs) stand out due to their flexibility and effectiveness in representing multivariate and non-linear relationships, having demonstrated robust performance in river water quality modeling under diverse physical, chemical, and hydrological conditions (Rahman and Chakrabarty, 2020). However, high-quality observational data are essential for this approach (Vercruysse et al., 2017); discharge and historical sediment records are often the most influential input variables (Adnan et al., 2019), and model performance depends on selecting appropriate predictors and ensuring a representative training dataset.

Because such in-situ datasets are often limited or unevenly distributed, remote sensing has become an important complementary data source for ML-based SSC modeling, offering a valuable pathway for large-scale applications and data-scarce regions (Peterson et al., 2018). Currently, satellites and remote sensors provide estimations of various hydrological parameters, including rainfall, evapotranspiration, turbidity, and water temperature (Fassoni-Andrade et al., 2021). With continuous advancements in sensor technology, the temporal and spatial resolution, as well as the accuracy of these data, have been improving, making remote sensing a powerful tool for SSC monitoring (Gebremicael et al., 2017). A traditional remote sensing-based method for estimating SSC involves the use of multispectral satellite imagery, as there is a well-established correlation between the concentration of suspended particles and the reflectance of solar radiation. As SSC increases, water bodies scatter more radiation, leading to higher reflectance in the water (Fassoni-Andrade and de Paiva, 2019). Despite its effectiveness, this method faces limitations, particularly cloud cover, which often hampers data acquisition during the very rainy periods when sediment transport is most critical.

To overcome this limitation of optical methods, remotely sensed precipitation data and global soil property datasets provide alternative pathways for SSC estimation. Because rainfall intensity (erosivity) and soil properties (erodibility) are primary drivers of erosion, they directly influence the amount of sediment transported into rivers. Linking these remotely accessible variables to SSC therefore establishes an indirect, but robust relationship between rainfall and soil (Vercruysse et al., 2017).

Despite the value of precipitation and soil datasets in explaining erosion-driven sediment transport, most studies that used remotely sensed variables to estimate SSC continue to rely primarily on surface reflectance data. Consequently, these approaches often overlook other hydrological variables, such as rainfall, soil properties, or flow conditions, that influence sediment dynamics (e.g., Aires et al., 2022; Duan et al., 2024; Kabir and Ahmari, 2020; Mollaee et al., 2018; Pereira et al., 2019; Peterson et al., 2018). Conversely, models that employ hydrometeorological variables—such as precipitation, discharge, soil erodibility, and evapotranspiration—typically do not integrate surface reflectance into the estimation framework (e.g., Aldin Shojaeezadeh et al., 2024; Lamane et al., 2025; Nguyen et al., 2024; Salih et al., 2020). Only a limited number of studies have combined both surface reflectance variables and hydrometeorological variables to estimate SSC (Fang et al., 2025). Therefore, the combined use of surface reflectance and hydrological predictors has been relatively underexplored, even though integrating this complementary information from both erosion drivers and optical water responses may improve model performance. Incorporating these complementary data sources has the potential to increase the accuracy and robustness of SSC estimation models, especially in regions with limited in situ data or frequent cloud cover.

Advances in remote sensing and the application of ML provide powerful tools to exploit such integrated datasets, as they are capable of capturing complex, non-linear relationships beyond the reach of traditional models. The growing availability of remote sensing data, when combined with ML techniques, enables the identification of patterns that conventional approaches often fail to capture. In particular, modeling the relationship between satellite-derived reflectance and SSC is challenging due to influences such as atmospheric interference, sun angle, and variations in water surface conditions (Zhang et al., 2012). ML techniques have shown greater ability to handle large and complex datasets, automatically extracting relevant features and improving predictive performance. These models offer advantages such as ease of implementation, fast computation, and high accuracy, making them increasingly common in the monitoring and management of water quality and sediment dynamics.

In this context, the objective of this study is to evaluate SSC estimation in a region with limited spatial and temporal in situ measurements by developing ANNs based on three distinct satellite-derived input configurations: (i) surface reflectance from multispectral imagery; (ii) precipitation and soil property data; and (iii) an integrated approach combining all variables. This framework enables the assessment of each input type's individual contribution and the potential benefits of merging them into a multi-source model. The ANN results are also compared with simpler traditional SSC rating-curves. The dataset used comprises reflectance from Landsat-8, Landsat-9, and Sentinel-2; daily precipitation from the CHIRPS product; and global soil property information.

2. Material and methods

Fig. 1 presents an overview of the methodological framework developed to estimate SSC using ANN models. In this schematic, blue boxes represent steps performed within Google Earth Engine (GEE), while orange boxes correspond to processes executed in MATLAB, particularly those related to ANN development. The workflow involves the selection of different input configurations, the acquisition and preprocessing of satellite and hydrological data, and subsequent model training and application. Three ANN configurations were tested using distinct input combinations: (i) surface reflectance derived from multispectral imagery (Sentinel-2 and Landsat-8/9), (ii) hydrological variables including daily precipitation (CHIRPS; Funk et al., 2015), soil properties (SoilGrids; Batjes et al., 2020), and drainage area (HydroSHEDS; Lehner and Grill, 2013), and (iii) an integrated approach combining all variables. These time series were matched with in situ SSC measurements from gauging stations for model calibration. Surface reflectance data underwent preprocessing steps such as spectral resampling, bad pixel and water masking, and sunglint correction. Hydrological variables were obtained by delineating watersheds at each station and extracting precipitation, soil, and drainage characteristics, resulting in a gridded dataset covering the period from 2013 to 2024. After assembling all input data, outliers were removed, and the matched records were used to train the ANN models. The trained models were

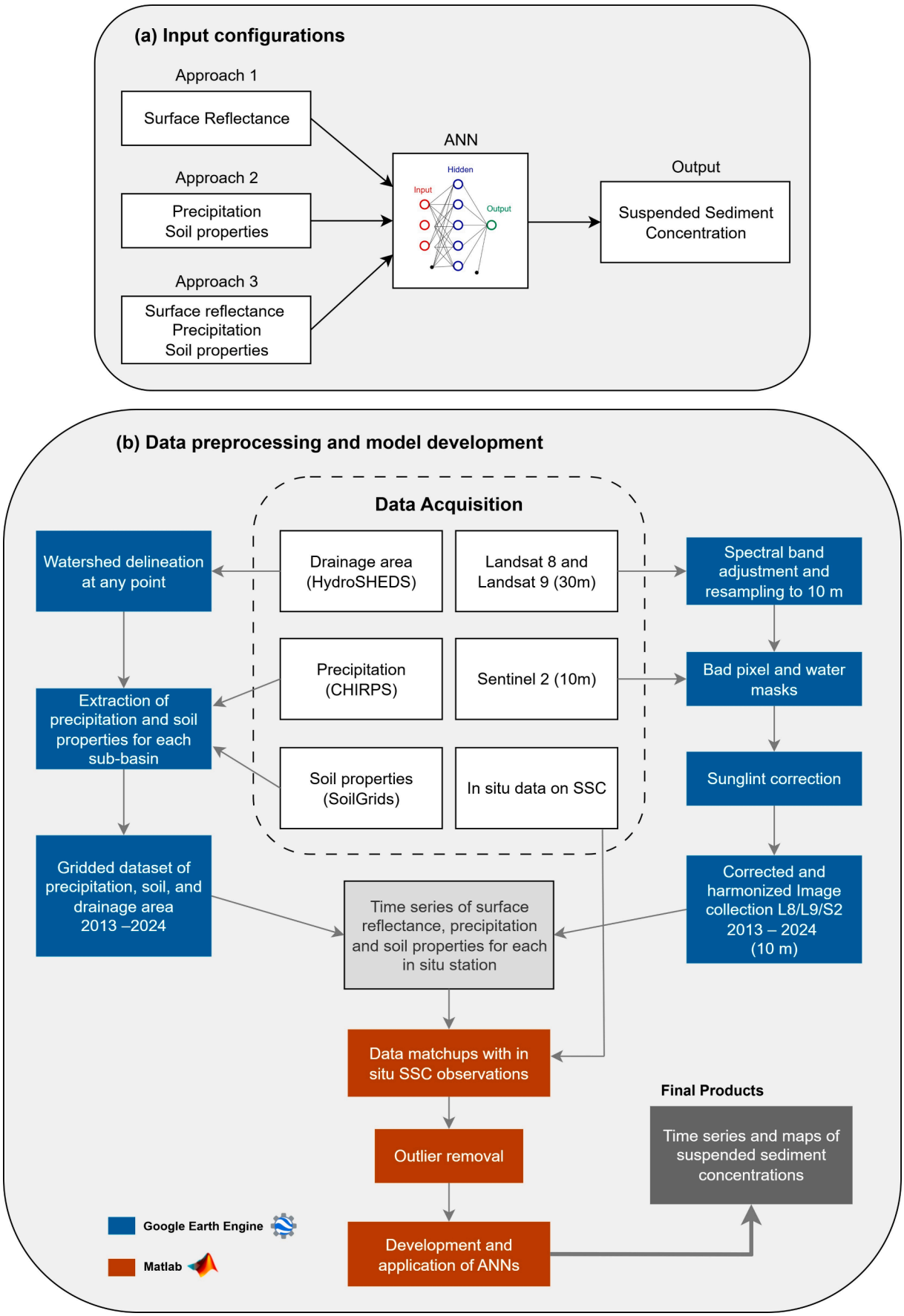


Fig. 1. Overview of the methodological framework for SSC estimation using remote sensing and ANNs. (a) Input configuration; and (b) data preprocessing and model development.

deployed in GEE to generate continuous time series and spatially explicit maps of SSC. The following subsections provide detailed descriptions of the study area, datasets, preprocessing methods, and ANN development.

2.1. Study area and in situ data

The study was carried out in two major river basins in southeastern Brazil: the Upper São Francisco River Basin and the Doce River Basin (Fig. 2). The Upper São Francisco Basin extends approximately between 14°10′–21°00′ S and 47°00′–42°30′ W and covers roughly 233,480 km^2. It lies predominantly within the Cerrado biome, a region characterized by heterogeneous vegetation formations ranging from grasslands to woodland savannas (Bettencourt et al., 2016). The Doce River Basin, located between about 17°45′–21°15′ S and 43°45′–39°30′ W, encompasses approximately 83,465 km2 and is largely situated within the Atlantic Forest biome, with remnants of native forest concentrated mainly in steeper terrain (Costa et al., 2022).

Climate in both basins exhibits marked seasonality. According to the Köppen climate classification, the Upper São Francisco Basin encompasses regions of Cwa—humid subtropical climate with dry winters and hot summers—and Cwb—temperate highland climate with dry winters and mild summers (Alvares et al., 2013). Annual precipitation in this basin decreases from approximately 1700 mm in the headwaters to about 1000 mm in downstream areas (Medeiros et al., 2024). The Doce River Basin also presents a combination of Köppen climate types, including Cwb in high-elevation zones, Cwa in the upper dissected terrain, and Aw, a tropical savanna climate with a pronounced dry winter, in the middle and lower basin (Alvares et al., 2013). In the Doce Basin, mean annual precipitation typically ranges from 1600 mm to over 2300 mm, with higher values occurring in mountainous areas (Domingues, 2011). Despite these spatial differences, both basins share a similar seasonal rainfall pattern: the November–January period contributes roughly 55–65% of the annual precipitation, followed by a well-defined dry season during austral autumn and winter.

These basins were selected because both were impacted by major mining tailings-dam failures that strongly influenced sediment dynamics. The Fundão dam failure near Mariana in 2015 discharged large volumes of sediment-rich waste into tributaries of the Doce, such as the Rio do Carmo, while the Córrego do Feijão dam failure near Brumadinho in 2019 heavily affected the Paraopeba River, a major tributary of the São Francisco. These extreme events provide valuable context for evaluating model performance under both typical and disturbed sediment conditions.

Hydrological and sediment data were compiled for 41 gauging stations across the basins (Table 1; locations in Fig. 2).

Station selection required river widths exceeding 30 m and the

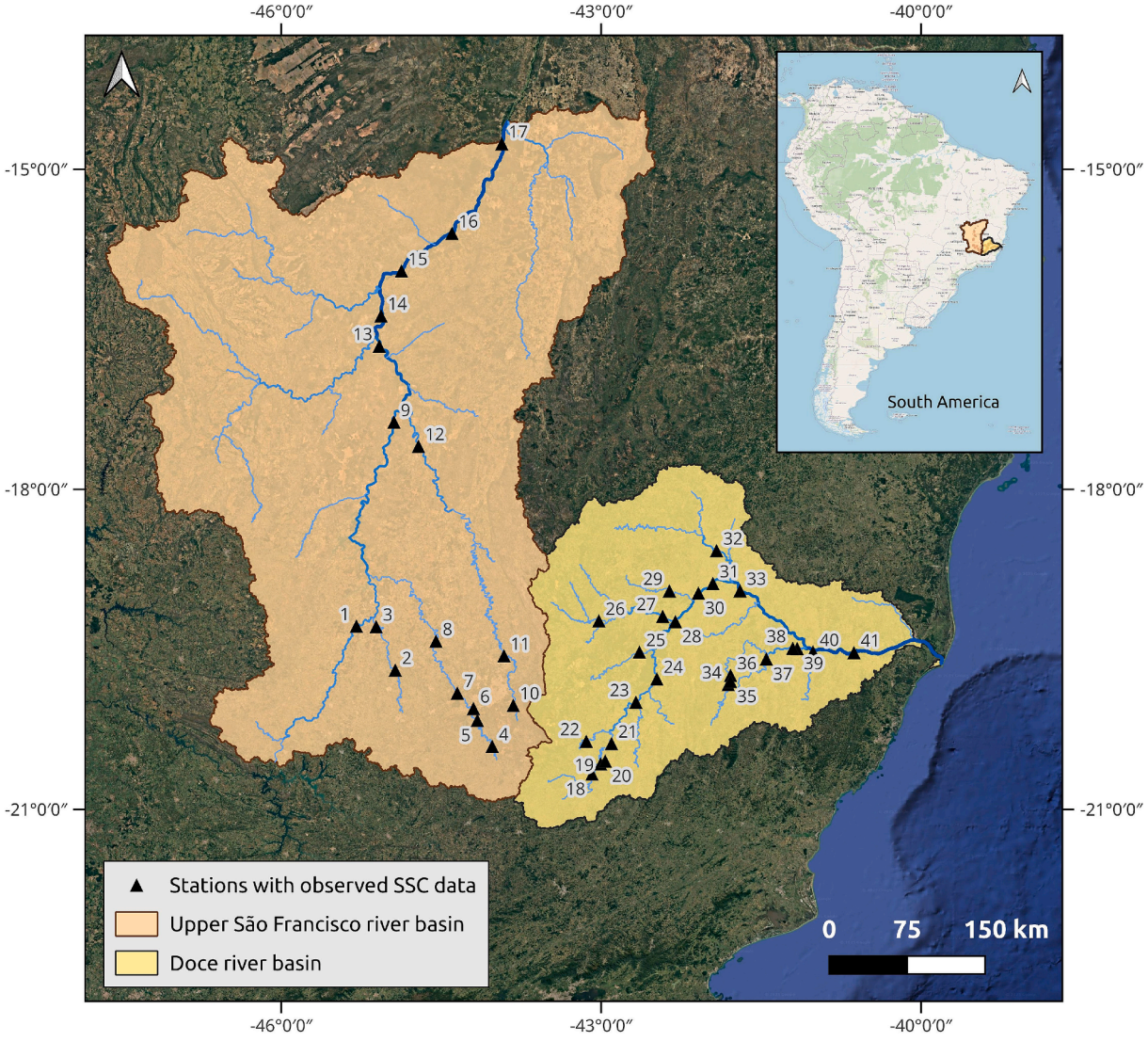


Fig. 2. Study area.

Table 1
In situ gauging stations.

ID	River	Sampling Period	Samples	Width (m)	Mean[a]	Std	Q50	Q10	Q90
1	São Francisco	2013–2023	22	60	91.7	124.3	38.2	10.7	369.8
2	Pará	2014–2023	20	55	18.1	18.9	7.0	2.6	48.0
3	Pará	2013–2018	13	80	46.9	53.4	28.2	9.9	133.4
4	Paraopeba	2013–2019	18	35	122.6	163.5	43.2	8.0	403.5
5	Paraopeba	2019–2020	18	40	147.8	173.3	54.9	8.9	464.8
6	Paraopeba	2019–2020	42	55	112.5	110.7	51.4	9.1	303.5
7	Paraopeba	2013–2022	71	70	114.0	135.6	67.2	9.6	326.3
8	Paraopeba	2019–2020	47	70	103.9	130.1	56.4	8.6	319.2
9	São Francisco	2013–2023	23	250	65.1	83.3	31.8	4.9	216.1
10	das Velhas	2013–2022	21	36	28.8	33.1	14.0	4.9	56.7
11	das Velhas	2013–2023	16	60	51.7	38.0	44.9	22.6	97.1
12	das Velhas	2013–2023	26	140	77.3	118.4	25.3	10.9	339.6
13	São Francisco	2013–2023	22	360	95.1	114.6	40.0	17.7	265.0
14	São Francisco	2013–2023	24	400	74.5	89.1	39.1	11.5	221.8
15	São Francisco	2013–2023	23	365	82.3	90.1	36.8	12.5	209.1
16	São Francisco	2013–2023	24	455	92.4	131.4	42.7	21.4	206.2
17	São Francisco	2013–2022	20	400	71.0	72.7	38.1	11.4	168.6
18	Piranga	2013–2022	26	35	59.0	146.2	11.1	3.6	137.6
19	Piranga	2013–2021	27	50	37.5	43.1	19.5	5.3	99.5
20	Piranga	2013–2021	28	90	34.3	68.4	11.9	4.0	52.4
21	Piranga	2013–2023	25	48	42.4	90.9	8.7	2.8	92.0
22	do Carmo	2013–2022	25	35	54.5	127.2	9.7	2.3	193.2
23	Doce	2013–2023	30	115	132.1	178.8	63.0	17.6	475.1
24	Doce	2013–2022	25	120	96.5	100.8	58.0	17.2	295.2
25	Piracicaba	2013–2023	27	50	33.6	47.6	11.7	5.3	106.2
26	Santo Antônio	2013–2022	19	55	45.1	122.9	5.6	2.1	95.3
27	Santo Antônio	2013–2023	25	135	42.2	55.9	16.9	8.2	132.0
28	Doce	2013–2020	32	180	50.5	64.1	26.5	8.2	99.9
29	Corrente Grande	2013–2022	23	35	43.9	80.1	11.9	3.9	130.5
30	Doce	2013–2020	32	240	28.2	33.8	12.0	3.9	73.5
31	Doce	2015–2023	19	300	99.9	146.5	27.1	6.3	369.2
32	Suaçui Grande	2013–2023	24	41	33.2	46.4	15.1	4.5	97.1
33	Doce	2013–2022	23	340	105.2	128.4	15.8	3.1	360.4
34	Manhuaçu	2013–2013	1	38	162	0	162	162	162
35	Manhuaçu	2013–2019	29	35	23.9	55.9	5.2	2.3	50.4
36	Manhuaçu	2013–2018	17	30	20.5	53.1	2.4	1.1	47.1
37	José Pedro	2013–2022	18	42	11.7	10.8	6.3	3.4	28.9
38	Manhuaçu	2015–2018	7	60	5.4	3.8	5.0	1.5	11.6
39	Manhuaçu	2013–2019	8	70	76.4	195.5	4.9	3.3	396.8
40	Guandu	2013–2022	20	32	28.0	29.5	21.2	3.6	72.7
41	Doce	2015–2023	18	588	104.6	169.5	25.5	6.1	415.6
Total		**2013–2023**	**978**		**79.4**	**128.3**	**25.9**	**4.4**	**233.4**

[a] Mean, Std, Q50, Q10, and Q90 refer to SSC values in mg L^{-1}.

availability of SSC observations. A total of 978 SSC samples were obtained from the Brazilian National Water Resources Information System (SNIRH – www.hidroweb.gov.br). Because individual stations contained limited SSC observations, station-specific models were not feasible. Instead, measurements from all stations were aggregated into a unified dataset to train and evaluate ANN models, enabling the models to capture spatial variability and learn nonlinear relationships between environmental drivers and sediment transport across both river systems.

2.2. Data

2.2.1. Multispectral images

We used multispectral satellite imagery from the Landsat 8 and Landsat 9 Operational Land Imager (OLI) sensors, as well as from the Sentinel-2 Multi-Spectral Instrument (MSI). Landsat 8 (launched in 2013) and Landsat 9 (launched in 2021), operated jointly by NASA and the U.S. Geological Survey (USGS), acquire imagery with a 30 m spatial resolution and a 16-day revisit cycle, reduced to 8 days when the two platforms are combined. We used Landsat Collection 2 Level-2 surface reflectance products processed with the Landsat Surface Reflectance Code (LaSRC) atmospheric correction algorithm (Vermote et al., 2018). Sentinel-2, operated by the European Space Agency (ESA) within the Copernicus program, consists of the Sentinel-2A (2015) and Sentinel-2B (2017) satellites, which provide 10 m resolution imagery and a combined 5-day revisit frequency. For Sentinel-2, we used Level-2A atmospherically corrected surface reflectance derived from the Sen2Cor processor (Muller-Wilm, 2012).

Image selection and pre-processing for Landsat 8/9 and Sentinel-2 were conducted on the Google Earth Engine (GEE) platform, a free, cloud-based platform for geospatial data analysis and visualization using JavaScript within its web code editor (https://code.earthengine.google.com/). GEE hosts an extensive archive of satellite imagery, including the datasets from Landsat 8/9 and Sentinel-2 used in this study. We selected all available images within the study area, filtered using a maximum cloud cover threshold of 50%, covering Landsat 8/9 Level-2 imagery from April 2013 to September 2024, and Sentinel-2 Level-2A imagery from December 2018 to September 2024, for subsequent analysis. In total, 8,699 images from L8/L9 and 42,366 images from Sentinel-2 were acquired seamlessly via GEE.

Landsat images were resampled to 10-meter spatial resolution to align with Sentinel imagery. Poor-quality pixels (clouds, cloud shadows, and saturated pixels) were masked using the following approach: (1) For Landsat images, we used the QA_PIXEL band, which includes pixel quality attributes generated using the CFMASK algorithm (Foga et al., 2017). Pixels flagged for clouds, cirrus, or cloud shadows were masked, and only water pixels were retained. (2) For Sentinel-2 images, the Scene Classification Map (SCL), Cloud Probability Map (MSK_CLDPRB), and QA60 bands were used to mask poor-quality pixels while retaining only those classified as water.

Some images affected by sunglint were corrected using a

straightforward method that subtracts the shortwave infrared (SWIR) band reflectance values from the visible and near-infrared bands on a pixel-by-pixel basis. This correction assumes that radiance from water in the SWIR range is negligible, and any detected signal in the SWIR band can be attributed to sunglint (Kay et al., 2009). Visual inspection confirmed the method's effectiveness, consistent with previous studies that have successfully applied this correction (Kuhn et al., 2019).

To obtain a representative reflectance value for each station and date, we extracted all pixels classified as water within a 500-meter radius around each gauging station (Fig. 3). This radius is used solely to define a search area from which valid water pixels are selected. Within this area, water pixels were identified using the MNDWI (Modified Normalized Difference Water Index) and the pixel-quality information from Sentinel-2 (Scene Classification Layer, SCL) and Landsat (QA band). Many of the analyzed rivers are relatively narrow, and individual pixels can be affected by noise from riverbanks, shadows, boats, or eventual residual atmospheric effects. By selecting only valid water pixels and computing the median reflectance of this group, we minimize the influence of such anomalous pixels and obtain a more stable and reliable reflectance estimate for each satellite overpass.

2.2.2. Precipitation data

Daily rainfall data were obtained from the CHIRPS (Climate Hazards Group InfraRed Precipitation with Station data) product (Funk et al., 2015), which provides quasi-global precipitation estimates at a spatial resolution of 0.05° (~5 km). CHIRPS combines satellite-based infrared observations with in situ station measurements to produce consistent and long-term rainfall time series, making it suitable for hydrological applications in regions with sparse monitoring networks. The dataset was accessed and processed through the Google Earth Engine (GEE) platform.

Rainfall data were extracted for each gauging station by averaging CHIRPS precipitation values over its upstream drainage area. Daily time series were produced for the entire study period, from January 1, 2013, to October 30, 2024. In addition to daily precipitation on the sampling day (t), several antecedent rainfall indicators were derived to capture short-term and longer-term hydrological responses. These included lagged daily precipitation values for the previous five days (t–1 to t–5) as well as cumulative rainfall over multiple antecedent windows (2, 3, 4, 5, 6, 7, 8, 9, 10, 15, and 30 days). These variables were incorporated to evaluate the influence of recent and accumulated rainfall on suspended sediment concentration dynamics.

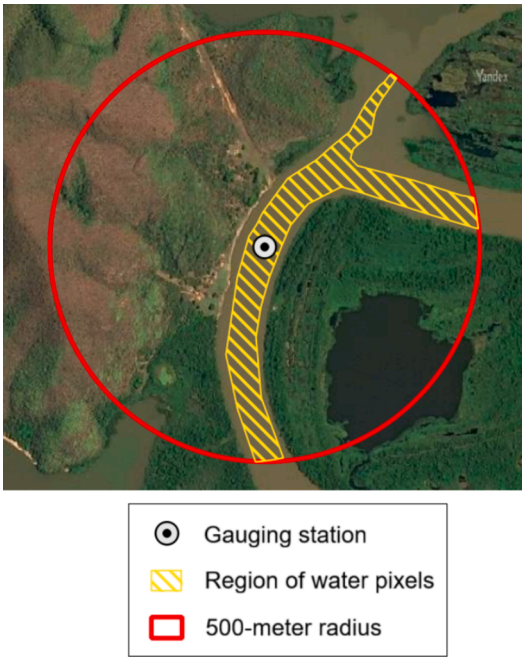


Fig. 3. Example of region of influence for gauging stations. The hatched area indicates the region where water pixels were used to calculate median reflectance values.

2.2.3. Soil properties

Soil properties were obtained from the SoilGrids product, a global digital soil mapping system developed and maintained by ISRIC – World Soil Information (Batjes et al., 2020). SoilGrids utilizes a global compilation of soil profile observations collected approximately between 1960 and 2020, combined with environmental covariates (e.g., climate, topography, vegetation indices), to generate globally consistent predictions of soil properties and classes at a spatial resolution of 250 m. The resulting maps are time-invariant, representing long-term average soil conditions. The dataset provides predictions for multiple soil depths and key properties relevant to hydrological and sediment transport modeling, including sand, silt, and clay content (g/kg), organic carbon, bulk density, and pH. For this study, sand, silt, and clay contents were extracted specifically for the surface soil layer (0–5 cm) to capture the primary interface influencing surface runoff and sediment detachment. Data extraction was conducted using the Google Earth Engine (GEE) platform, which provides seamless and open access to SoilGrids, enabling efficient retrieval of soil property data pixel by pixel across the drainage areas of each gauging station.

2.2.4. Watershed delineation

For extracting precipitation and soil property data, it was necessary to delineate the upstream drainage areas of the in situ gauging stations. This was performed in Google Earth Engine (GEE) using the HydroBASINS database (Lehner and Grill, 2013), which provides a hierarchical subdivision of global river basins into 11 spatial levels. In this study, Level 7 was selected because it provides a suitable balance between spatial detail and computational efficiency for the watershed sizes in the study region. Lower levels would result in overly coarse basin boundaries, whereas higher levels would increase the number of subdivisions and computational cost without improving delineation accuracy. Thus, Level 7 offered the most appropriate resolution for defining the contributing areas of the gauging stations.

2.2.5. Data filtering

Before modeling SSC, it was necessary to apply an outlier-removal procedure to both the multispectral reflectance time series and the in situ SSC measurements. Although the remote sensing datasets had undergone standard preprocessing—such as cloud masking, quality filtering, and visual inspection—residual anomalous values remained, particularly associated with incomplete cloud detection, atmospheric contamination or sensor noise.

For the multispectral reflectance time series, the outlier-removal procedure operates by identifying reflectance values that depart markedly from the expected temporal behavior of the signal. As illustrated in Fig. 4, the original reflectance series (black line) displays abrupt spikes and dips that do not follow the smoother seasonal pattern represented by the moving-average curve (blue line). These abrupt deviations often arise from residual contamination—such as clouds, haze, atmospheric scattering, or sensor noise—even after standard preprocessing. To detect such anomalies, we computed a moving average over a short temporal window (5 days for Landsat-8/9 and 7 days for Sentinel-2), assuming that true surface reflectance varies gradually in response to hydrological seasonality. We then quantified the difference between each observation and its local moving average (magenta line in Fig. 4). Under the assumption that these differences follow an approximately symmetric distribution around zero, we defined statistical thresholds corresponding to the 95th and 5th percentiles (red lines). Observations falling outside this interval represent unusually large deviations from their temporal neighbors and were therefore classified as outliers and

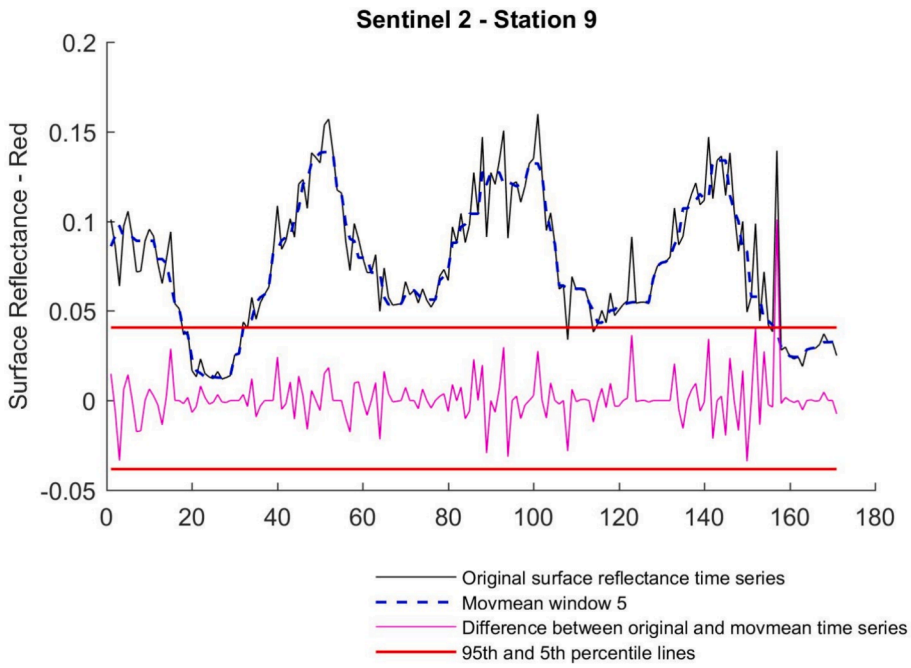


Fig. 4. Example of the outlier-removal procedure applied to a surface-reflectance time series.

removed.

For the in situ SSC measurements, outliers were manually removed to eliminate values influenced by exceptional or non-representative conditions. Measurements collected during the period surrounding the failures of mining tailings dams—events that released large volumes of sediment-rich waste material into the river system—as well as those taken in downstream reaches affected by these disturbances, were excluded because they represent extreme sediment pulses unrelated to natural fluvial behavior. Additionally, samples flagged by the national water agency—the institution responsible for collecting and providing the in situ data—as containing excessive coarse sand were removed. This condition indicates accidental inclusion of bed material during sampling, which does not represent true suspended sediment and would bias the dataset. However, although these values were excluded from the model calibration, the dam-failure-related observations were reintroduced in the final analysis to enable a visual comparison between the simulated results and the extreme measured SSC values.

2.3. Input variable selection

A comprehensive set of input variables was used in this study to support the modeling of suspended sediment concentration (Table 2). The spectral indices listed as inputs 4 to 13 in Table 2 were derived from combinations of visible and near-infrared bands (Red, Green, NIR) to enhance optical contrasts related to water quality. These indices capture different aspects of water-leaving reflectance, particularly variations associated with suspended sediment concentration and turbidity. By exploring ratios and normalized differences between bands, they provide complementary information that improves the sensitivity of ANN models to sediment dynamics under diverse optical conditions.

2.4. Artificial neural networks (ANN)

Multilayer perceptron ANN models were developed with a three-layer architecture comprising an input layer, a hidden layer, and an

Table 2
Selected input variables.

ID	Input	ID	Input	ID	Input	ID	Input
1	R green	11	(Green + Red)/2	21	Accum 3d	30	Accum 30d
2	R red	12	Red/(Green + NIR)	22	Accum 4d	31	Sand (g/kg)
3	R NIR	13	(Green − NIR)/(Green + NIR)	23	Accum 5d	32	Silt (g/kg)
4	(Red − Green)/(Red + Green)	14	Daily Precip	24	Accum 6d	33	Clay (g/kg)
5	(NIR − Red)/(NIR + Red)	15	Precip t−1	25	Accum 7d	34	Drainage area (km²)
6	Green/(Red + NIR)	16	Precip t−2	26	Accum 8d		
7	(Green − NIR)/(Red + NIR)	17	Precip t−3	27	Accum 9d		
8	Green/Red	18	Precip t−4	28	Accum 10d		
9	Red/NIR	19	Precip t−5	29	Accum 15d		
10	Red/Green	20	Accum 2d				

Spectral Reflectance Indices
Precipitation Variables from CHIRPS
Soil and Drainage Characteristics

output layer. Each layer utilized a unipolar sigmoid activation function. The input features included different combinations of remote sensing datasets and the output feature was in situ SSC. Initially, models were developed using only variables derived from multispectral satellite imagery to assess the predictive capacity of spectral information alone. Additional models were developed using only precipitation, soil, and watershed characteristics to evaluate their isolated predictive power. Finally, combined models were created, integrating all available variables to capture the potential synergies among spectral, rainfall, soil, and drainage characteristics for suspended sediment estimation.

Given the initial set of 34 input variables, an exploratory assessment of the spectral and hydrometeorological predictors was first conducted to characterize their relationships with SSC and to identify potential redundancies. Based on this assessment, a variable selection procedure was implemented to reduce model dimensionality while retaining predictive performance. This selection process was based on the inter-correlation between input variables, where pairs of highly correlated variables were identified, and only the variable with the strongest correlation with SSC was retained for modeling. Different exclusion criteria were tested, using inter-correlation thresholds of >0.9, 0.7, and 0.5, resulting in models with varying combinations and numbers of input variables. This systematic input reduction allowed for the development of optimized ANN models with reduced complexity while maintaining robust performance in predicting suspended sediment concentrations across different river systems.

The ANN models were structured as multilayer perceptrons with a three-layer architecture (input, hidden, and output layers) and a unipolar sigmoid activation function. To enhance prediction reliability, each model consisted of five ANNs with identical architecture but trained using different partitions of the dataset. The full set of remote sensing–SSC matchups was divided into five equal folds (20% each), all containing observations from the 41 gauging stations. In each rotation, three folds were used for training, one for validation, and one for independent testing, as illustrated in Fig. 5; the final SSC estimate was obtained by averaging the outputs of the five networks. The validation subset was used exclusively to monitor generalization and apply early stopping, whereas the test subset remained completely unseen during training. To ensure that the ANNs learned the full range of SSC variability, the minimum and maximum input–output values were always assigned to the training set, allowing the models to capture both typical conditions and the highest sediment concentrations.

During model training, the connection weights and biases of each ANN were adjusted through the backpropagation procedure (Rumelhart et al., 1986), using only the training portion of the data. To reduce the risk of overfitting, a validation-based early-stopping strategy was adopted. In this approach, training was interrupted once the validation error began to increase while the training error continued to decrease, which signals that the network is starting to memorize the training data rather than generalizing well (Hecht-Nielsen, 1990). The size of the hidden layer was defined through iterative experimentation, choosing the smallest number of neurons that still yielded stable and accurate results according to the validation performance (Sari et al., 2017). The performance of the calibrated ANNs was evaluated using three statistical metrics. The mean absolute error (MAE) was computed to quantify the average magnitude of the absolute differences between measured and predicted SSC values. The root mean square error (RMSE) was also calculated, providing a measure that emphasizes larger deviations between observations and model estimates. In addition, the coefficient of determination (R^2) was used to indicate how much of the variability in the observed SSC data was captured by the ANN predictions.

After training, the ANN functions were converted from MATLAB to JavaScript for implementation within Google Earth Engine. This enabled the application of the trained models directly to Landsat 8/9 and Sentinel-2 satellite imagery, allowing for the estimation of suspended sediment concentrations across all water pixels in the study regions.

2.5. SSC rating-curve

Some in situ SSC monitoring stations also provided concomitant river discharge measurements. Therefore, we selected four locations to fit traditional SSC rating-curves (Asselman, 2000) and compare their performance with the ANN models, assessing how SSC estimates derived from remote-sensing variables perform relative to discharge-only estimates. The rating-curves were adjusted using a power-law function, as expressed below:

$$SSC = a^{*}Q^{b}$$

Where: SSC – suspended sediment concentration; Q – river discharge; and a and b – calibration coefficients.

After developing both the ANN models and the SSC rating-curves, the models were grouped and analyzed according to three categories: (i) models based on surface-reflectance indices; (ii) models based on precipitation and soil properties; and(iii) models combining surface reflectance, precipitation, and soil properties.

Results were presented using SSC time-series plots for each location, scatterplots of observed versus estimated values, and quarterly mean maps of suspended sediment concentrations over water-identified pixels. The four stations selected for comparison (Stations 7, 16, 23, and 41) represent contrasting hydrological and geomorphological

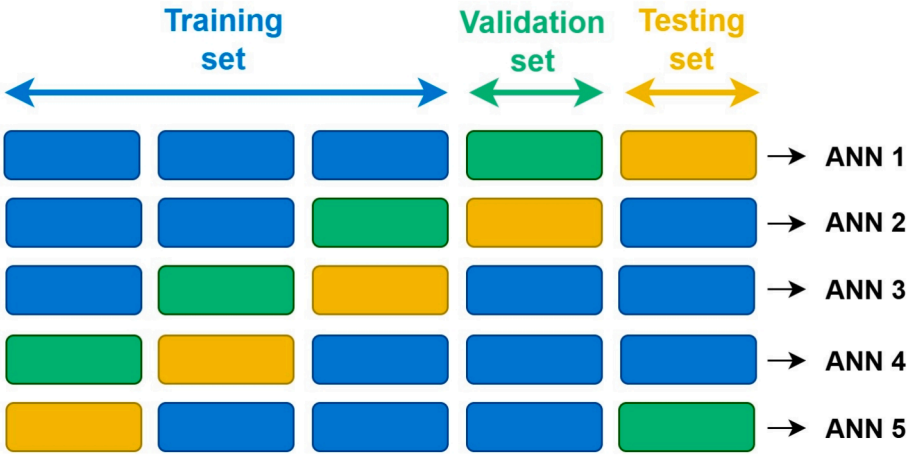


Fig. 5. The subdivision of the dataset into training (blue boxes), validation (green boxes), and testing (yellow boxes) for the development of the ANNs. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

conditions across the two river basins analyzed in this study. Additionally, Stations 7 and 23 correspond to smaller river reaches, whereas Stations 16 and 41 are located along larger river stretches within each basin.

3. Results

3.1. Exploratory assessment of predictors

The exploratory assessment of the spectral and hydrological predictors highlighted clear patterns in their relationship with SSC. Fig. 6a shows that variables derived from combinations of the green and red reflectance bands, together with the 15-day and 30-day cumulative rainfall, exhibit the strongest associations with SSC, confirming their relevance for sediment estimation. These results reinforce previous evidence that integrating rainfall and soil-related information helps reduce temporal and spatial scale mismatches inherent in remote-sensing SSC retrievals (Mohamadi and Kavian, 2015; Vercruysse et al., 2017). Complementary to this, the inter-variable correlation structure (Fig. 6b) revealed pronounced redundancy among several inputs, particularly among spectral indices of similar composition and among precipitation metrics with different accumulation windows. These clusters of highly correlated variables provided quantitative support for implementing input-reduction strategies during model development. In practice, variables exceeding the predefined correlation thresholds were progressively removed, prioritizing those most strongly associated with SSC. This step ensured a more parsimonious and interpretable predictor set while preserving variables with the greatest predictive potential.

3.2. Performance of the ANN models

After the input reduction strategy was employed, several ANN models were developed. Table 3 summarizes the performance of the three groups of ANN models.

Models based solely on surface reflectance (ANN1–ANN5) showed strong performance during training, with R^2 between 0.83 and 0.87, MAE ranging from 20.3 to 21.9 mg/L, and RMSE between 30.4 and 33.9 mg/L. In the testing set, performance diverged within this group: ANN1 and ANN2 showed reduced generalization ($R^2 = 0.59$, MAE ≈ 29–30 mg/L, RMSE ≈ 52–53 mg/L), whereas ANN3, ANN4, and ANN5 maintained high predictive skill, with R^2 values of 0.79–0.81, MAE around 21.9–22.9 mg/L, and RMSE between 33.9 and 39.2 mg/L. These results indicate that, when appropriately regularized and with redundant inputs removed, reflectance-only models can estimate SSC with good accuracy, consistent with previous studies in the Doce River basin (Aires et al., 2022).

Models using only precipitation and soil properties (ANN6–ANN10) performed with less accuracy, reflecting the more indirect relationship between hydrometeorological forcing and SSC. Training R^2 values ranged from 0.65 to 0.80, with MAE between 17.8 and 27.4 mg/L and RMSE between 26.7 and 50.1 mg/L. In the testing set, R^2 dropped to 0.43–0.74, MAE increased to 26.7–34.0 mg/L, and RMSE rose to 43.2–68.9 mg/L. Within this group, ANN6 stands out as the best model (testing $R^2 = 0.74$, MAE = 26.7 mg/L, RMSE = 43.2 mg/L), showing that rainfall and watershed properties can provide reasonable SSC estimates when optical data are not available, even though they cannot capture event-driven sediment pulses unrelated to precipitation.

Hybrid models combining reflectance, precipitation, and soil variables (ANN11–ANN14) achieved the best overall performance. In the training set, these models reached R^2 values between 0.87 and 0.93, with MAE of 15.5–22.0 mg/L and RMSE of 22.8–30.1 mg/L. During testing, these models also generalized well: testing R^2 values ranged from 0.75 to 0.84, MAE from 18.7 to 24.3 mg/L, and RMSE from 29.5 to 33.7 mg/L. ANN13 was the best overall model according to the statistics, with testing $R^2 = 0.84$, MAE = 18.7 mg/L, and RMSE = 29.5 mg/L. By integrating complementary information from surface reflectance, precipitation, and soil properties, ANN13 best captures both seasonal SSC variability and short-lived sediment pulses, achieving accuracy comparable to more complex models that rely on a much larger set of predictors (e.g., Aires et al., 2023), but with fewer input variables and lower data requirements.

3.3. Temporal dynamics of SSC

According to the performance statistics, ANN4, ANN6, and ANN13 were selected as representative models of each architecture family. Figs. 7 and 8 present the SSC time series estimated by these models for the four selected gauging stations. A clear seasonal pattern is observed across all sites, with the lowest SSC concentrations occurring during the dry months (May–September) and the highest values during the rainy season (October–March). On top of this expected seasonal pattern, some periods show unusually high observed in situ SSC (red circles) caused by tailings-dam failures. At Station 7 (Fig. 7a–c), SSC rises sharply in early 2019 due to the Brumadinho tailings-dam collapse, which also triggered intensified in situ SSC monitoring that year. In the Doce River basin, Stations 23 and 41 (Fig. 8a–c and g–i) show elevated SSC immediately after November 2015, following the Fundão dam failure, with values remaining above typical seasonal levels during parts of 2016–2018.

At both Station 7 and Station 23, the reflectance-based models ANN4 and ANN13 captured some of the SSC pulses related to the Brumadinho and Fundão events, in some cases retrieving very high SSC values (exceeding 750 mg/L at Station 7). However, many of the highest in situ SSC values occurred during the rainy season, when cloud cover frequently prevented the acquisition of valid optical imagery. Consequently, some extreme SSC events were not predicted by ANN4 or ANN13 due to the absence of usable reflectance data. In contrast, the precipitation-based ANN6, which relies exclusively on hydrometeorological drivers, also failed to reproduce these extreme SSC values because they were not linked to rainfall anomalies but were instead directly triggered by tailings-dam failures that introduced large volumes of sediment into the Paraopeba River. As a result, some isolated high SSC observations remain unmatched by all models: for ANN4 and ANN13, due to cloud-induced data gaps, and for ANN6, because these events were not linked to rainfall.

At Station 16 (Fig. 7g–i), all three models exhibit consistent agreement. This is expected because this section of the São Francisco River is wide (>400 m) and drains a very large contributing area (>200,000 km^2), resulting in attenuated and slower hydrological responses. Moreover, this reach is located downstream of the Três Marias Dam and was not substantially affected by the Brumadinho tailings-dam failure. In large river systems, such the stretch of Station 16 at São Francisco river, hydrological responses tend to occur more slowly due to their greater channel width, storage capacity, and extended water and sediment travel times. These characteristics minimize the influence of short-duration rainfall events, resulting in smoother and more stable sediment-transport dynamics relative to smaller, more responsive catchments (Vercruysse et al., 2017). Therefore, large river systems tend to be more predictable, even when using models based solely on hydrometeorological inputs. At Station 41 (Fig. 8g–i), ANN4 and ANN13 again show sensitivity to atypically high SSC values, whereas ANN6 does not, although none of the models fully captures the magnitude of the SSC peaks observed during 2016–2018.

Figs. 7 and 8 (panels d–f and j–l) provide detailed zoomed-in views of the period from January 2018 to January 2020 for all four stations. While the larger panels illustrate the full seasonal cycles between 2013 and 2024, the zoomed-in panels allow for closer inspection of model behavior under low and moderate SSC conditions. Across all stations, the reflectance-based models generally show better performance in reproducing lower SSC values, although ANN13 exhibits a slightly larger uncertainty range due to variability propagated through its k-fold simulations. Although ANN6 provides daily SSC estimates, its accuracy is typically lower than that of models incorporating surface reflectance.

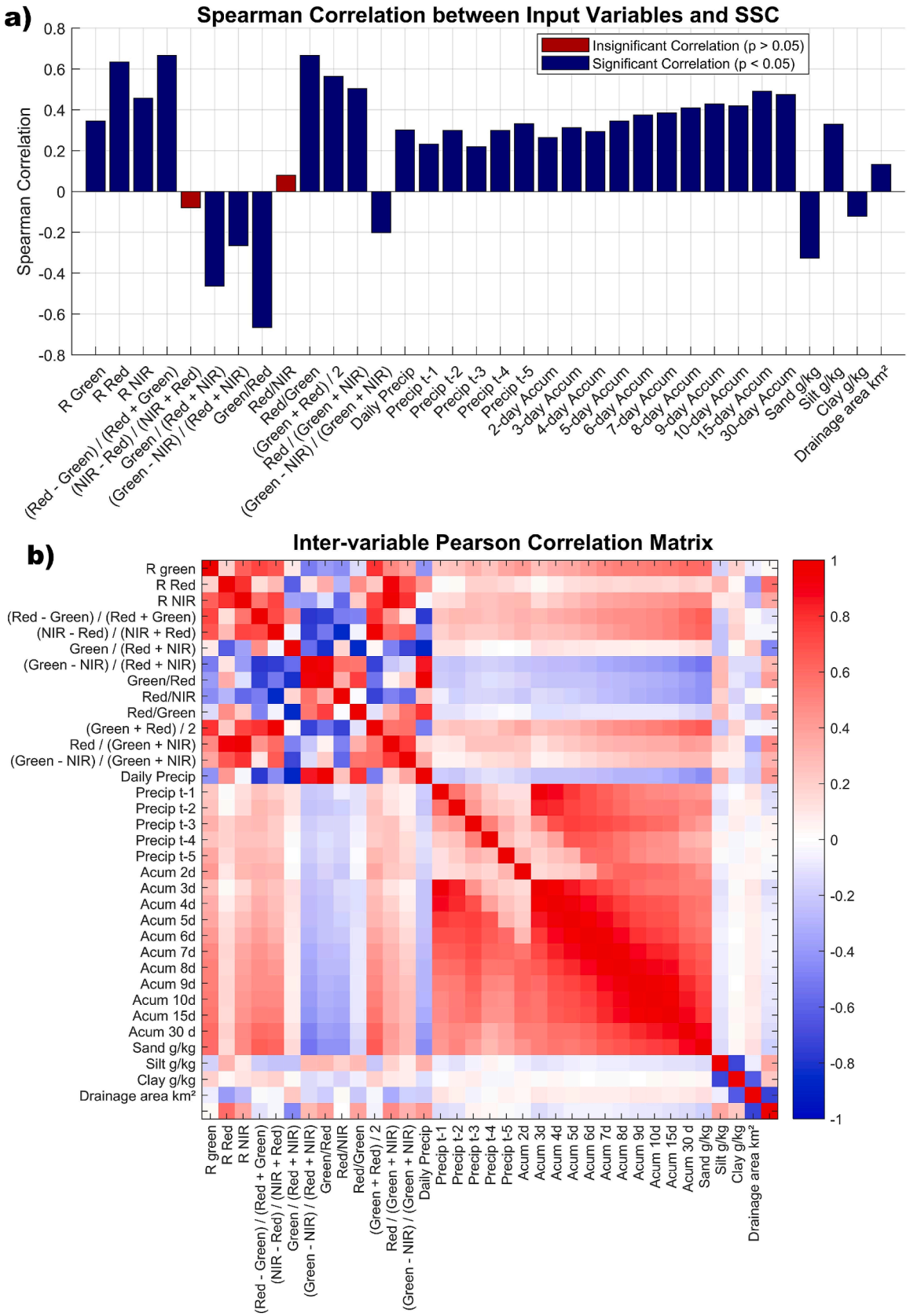


Fig. 6. (a) Spearman correlation coefficients between each input variable and suspended sediment concentration (SSC). Blue bars indicate statistically significant correlations ($p < 0.05$), (b) Inter-variable Pearson correlation matrix for the input variables used in the SSC modeling. The color scale ranges from −1 (strong negative correlation, blue) to +1 (strong positive correlation, red), with white indicating no correlation. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 3
Summary of the evaluated ANN models and their performance.

Models based on surface reflectance indexes						
Model	*Criteria for input selection*	*ID input variables*	*NHL*[a]	*Training R^2 / MAE / RMSE*	*Validation R^2 / MAE / RMSE*	*Testing R^2 / MAE / RMSE*
ANN1	All surface reflectance variables	1 to 13	7	0.83 / 21.9 / 33.9	0.76 / 28.1 / 46.7	0.59 / 30.0 / 52.7
ANN2	Inter correlation < 0.9	2, 3, 5, 6, 7, 8, 9, 10, 11 and 12	6	0.86 / 20.6 / 30.4	0.75 / 26.0 / 42.2	0.59 / 29.2 / 52.9
ANN3	Inter correlation < 0.7	2, 6, 7, 10 and 12	5	0.87 / 20.3 / 30.9	0.82 / 21.1 / 32.2	0.80 / 21.9 / 33.9
ANN4	Inter correlation < 0.5	2, 6, 10 and 12	5	0.87 / 21.1 / 32.1	0.83 / 21.3 / 32.3	0.81 / 22.9 / 36.7
ANN5	Simple model	1, 2, 3	4	0.87 / 20.4 / 33.4	0.82 / 20.9 / 33.1	0.79 / 22.4 / 39.2
Models based on precipitation and soil properties						
Model	*Criteria for input selection*	*ID input variables*	*NHL*[a]	*Training R^2 / MAE / RMSE*	*Validation R^2 / MAE / RMSE*	*Testing R^2 / MAE / RMSE*
ANN6	All precipitation and soil variables	14 to 33	8	0.80 / 17.8 / 26.7	0.78 / 24.2 / 40.5	0.74 / 26.7 / 43.2
ANN7	All Precipitation variables	14 to 30	7	0.78 / 23.3 / 33.5	0.75 / 25.3 / 42.4	0.69 / 27.8 / 50.6
ANN8	Inter correlation < 0.9	14, 15, 16, 17, 18, 19, 21, 23, 24, 25, 26, 27, 29, 30, 31, 32, 33, 34	7	0.65 / 27.4 / 50.1	0.55 / 29.2 / 57.7	0.43 / 34.0 / 68.9
ANN9	Inter correlation < 0.7	17, 18, 19, 21, 24, 25, 27, 29, 32 and 34	6	0.67 / 26.7 / 48.9	0.48 / 27.8 / 62.3	0.43 / 29.3 / 66.2
ANN10	Inter correlation < 0.5	25, 26, 27, 29, 32 and 34	5	0.76 / 25.4 / 40.6	0.60 / 27.6 / 59.1	0.44 / 30.3 / 66.9
Models based on surface reflectance, precipitation and soil properties						
Model	*Criteria for input selection*	*ID input variables*	*NHL*[a]	*Training R^2 / MAE / RMSE*	*Validation R^2 / MAE / RMSE*	*Testing R^2 / MAE / RMSE*
ANN11	All variables	1 to 34	8	0.87 / 22.0 / 30.1	0.75 / 23.8 / 30.7	0.75 / 24.3 / 33.7
ANN12	Inter correlation < 0.9	2, 3, 5, 6, 7, 8, 9, 10, 11, 12, 14, 15, 16, 17, 18, 19, 21,23, 24, 25, 26, 27, 29, 30, 31, 32, 33 and 34	7	0.89 / 18.2 / 26.3	0.82 / 24.7 / 30.2	0.73 / 19.7 / 30.1
ANN13	Inter correlation < 0.7	2, 6, 7, 10, 12, 17, 18, 19, 21, 24, 25, 27, 29, 32 and 34	7	0.93 / 15.5 / 22.8	0.88 / 16.8 / 25.6	0.84 / 18.7 / 29.5
ANN14	Inter correlation < 0.5	2, 6, 10, 12, 15, 26, 27, 29, 32 and 34	6	0.93 / 15.7 / 24.1	0.85 / 19.6 / 27.1	0.81 / 20.9 / 30.2

[a] NHL – Neurons in hidden layer.

In examining the zoomed-in period, it also becomes evident that the availability of in situ SSC data is extremely limited. Except for Station 7, the remaining monitoring stations contain six or fewer SSC observations within this two-year window, underscoring the pronounced scarcity of field measurements in the region and the challenge this poses for model validation.

3.4. Traditional SSC rating-curves vs. ANN models

Fig. 9 presents the SSC rating-curves derived for the four stations with concomitant discharge and SSC measurements. The fitted power-law relationships exhibit low to moderate explanatory power (R^2 = 0.16–0.59) and substantial scatter, highlighting the well-known limitations of rating-curve approaches for estimating suspended sediment concentrations. Numerous studies have shown that SSC is frequently asynchronous with discharge, producing rising- and falling-limb differences and a marked hysteresis, which occurs when the same discharge corresponds to different sediment concentrations (Javed et al., 2021; Jing et al., 2025). As a result, SSC rating-curves regularly underestimate sediment transport, often by 10% to more than 50% (Asselman, 2000). Even though such curves can help describe the general nonlinear association between SSC and discharge during storm events (Yu et al., 2023), they provide only a partial representation of the processes governing sediment mobilization. Importantly, however, SSC–Q hysteresis analysis remains valuable not for estimating SSC itself, but for diagnosing sediment sources and interpreting short-term sediment dynamics in watersheds (Cao et al., 2021; Lloyd et al., 2016).

Fig. 10 compares observed SSC with the estimates produced by the SSC rating-curves and the main ANN models (ANN4, ANN6, and ANN13) at the four stations where discharge and SSC measurements are available.

Consistent with the limited predictive skill observed in the SSC rating-curves (Fig. 9), the discharge-based estimates show low accuracy, with substantial dispersion around the 1:1 line, high errors (MAE and RMSE), and particularly weak performance at Stations 23 and 41, where the relationship between discharge and SSC is especially nonlinear and variable. In contrast, all ANN models provide considerably improved predictions across the four sites, with higher R^2 values and markedly lower error metrics. Notably, ANN6—despite relying solely on precipitation and soil variables—outperforms the traditional SSC rating-curves at all stations, indicating that rainfall-driven information can be as important as discharge for predicting sediment concentrations in river systems. ANN13 achieves the best overall performance, often capturing both low and high SSC values more consistently than ANN4 and ANN6.

These results also demonstrate that multisource ANNs, integrating multispectral reflectance, precipitation, and soil information, are able to represent the complex controls on sediment transport far better than traditional discharge-only rating curves. The clear performance gains highlight the potential of remote-sensing-based ANN models to support SSC estimation in data-scarce regions where in situ measurements are sparse and temporally limited.

3.5. Spatial and seasonal variability of SSC from reflectance-based ANN models

Reflectance-based models such as ANN4 and ANN13 allow the construction of spatially resolved SSC maps along river corridors, supporting detailed, pixel-level diagnostics. Fig. 11 illustrates the quarterly mean SSC estimates generated by ANN13 for the four gauging stations, highlighting the seasonal dynamics SSC across the Paraopeba, São Francisco, and Doce rivers. The maps reveal a consistent pattern of higher SSC values during the wet season (October–March) and lower concentrations in the dry season (April–September), reflecting rainfall-driven sediment mobilization in these basins.

The Paraopeba and Doce rivers exhibit their highest quarterly SSC values during October–December, whereas in the São Francisco River the maximum SSC occurs one quarter later, in January–March. Conversely, the lowest SSC values in the Paraopeba and Doce rivers appear in April–June, while in the São Francisco River the minimum concentrations occur during July–September. This reveals a one-quarter shift in hydrosedimentary seasonality between Station 16 in the São Francisco River and the other evaluated reaches. This offset is consistent

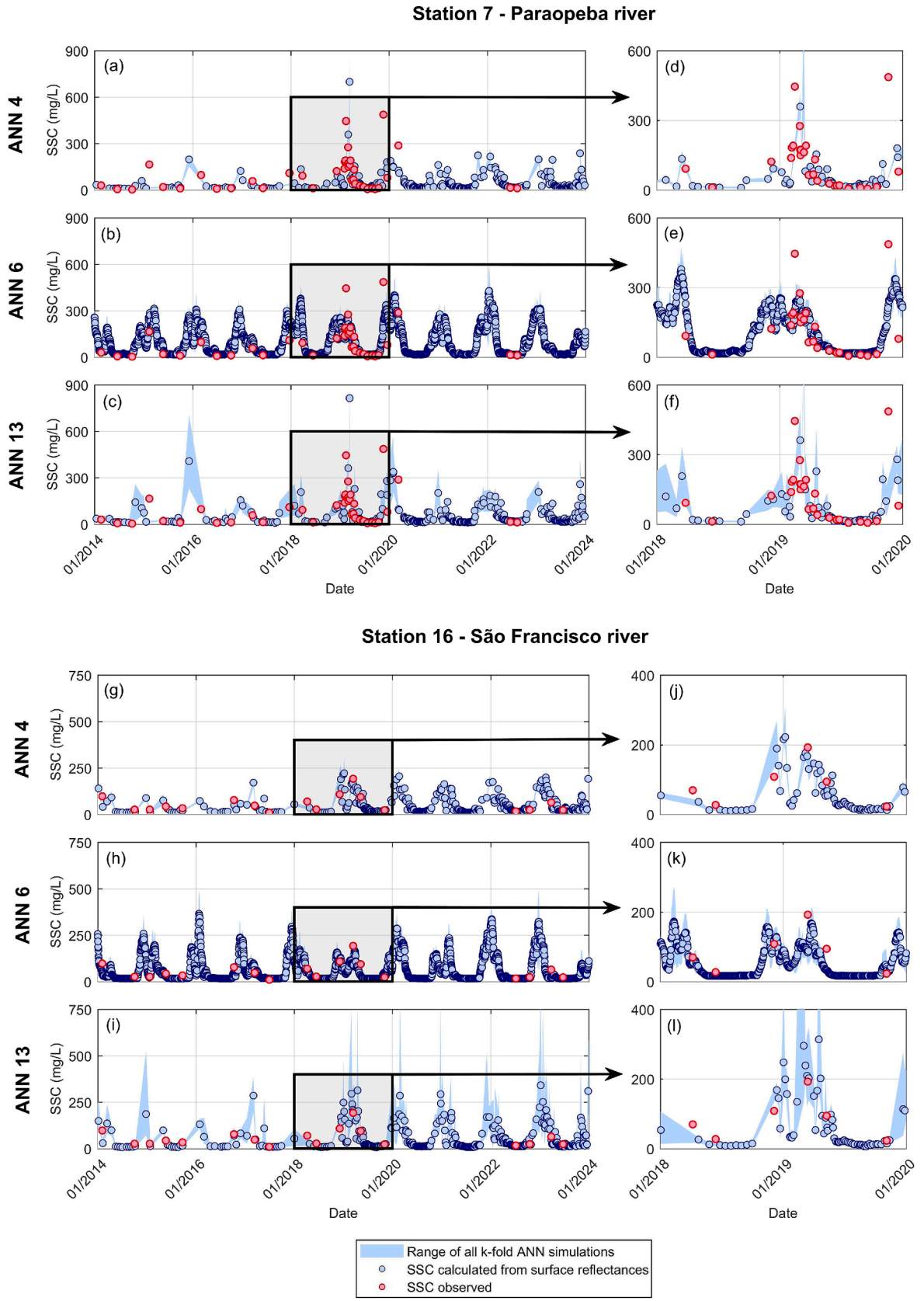


Fig. 7. SSC time series at Station 7 (Paraopeba River) and Station 16 (São Francisco River) using ANN4, ANN6, and ANN13. Panels (a)–(c) correspond to Station 7 and (g)–(i) to Station 16. Blue circles show SSC estimates, red circles denote observed values, and the blue shaded areas indicate the k-fold simulation range. Zoomed-in views of the boxed periods are presented in (d)–(f) for Station 7 and (j)–(l) for Station 16. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

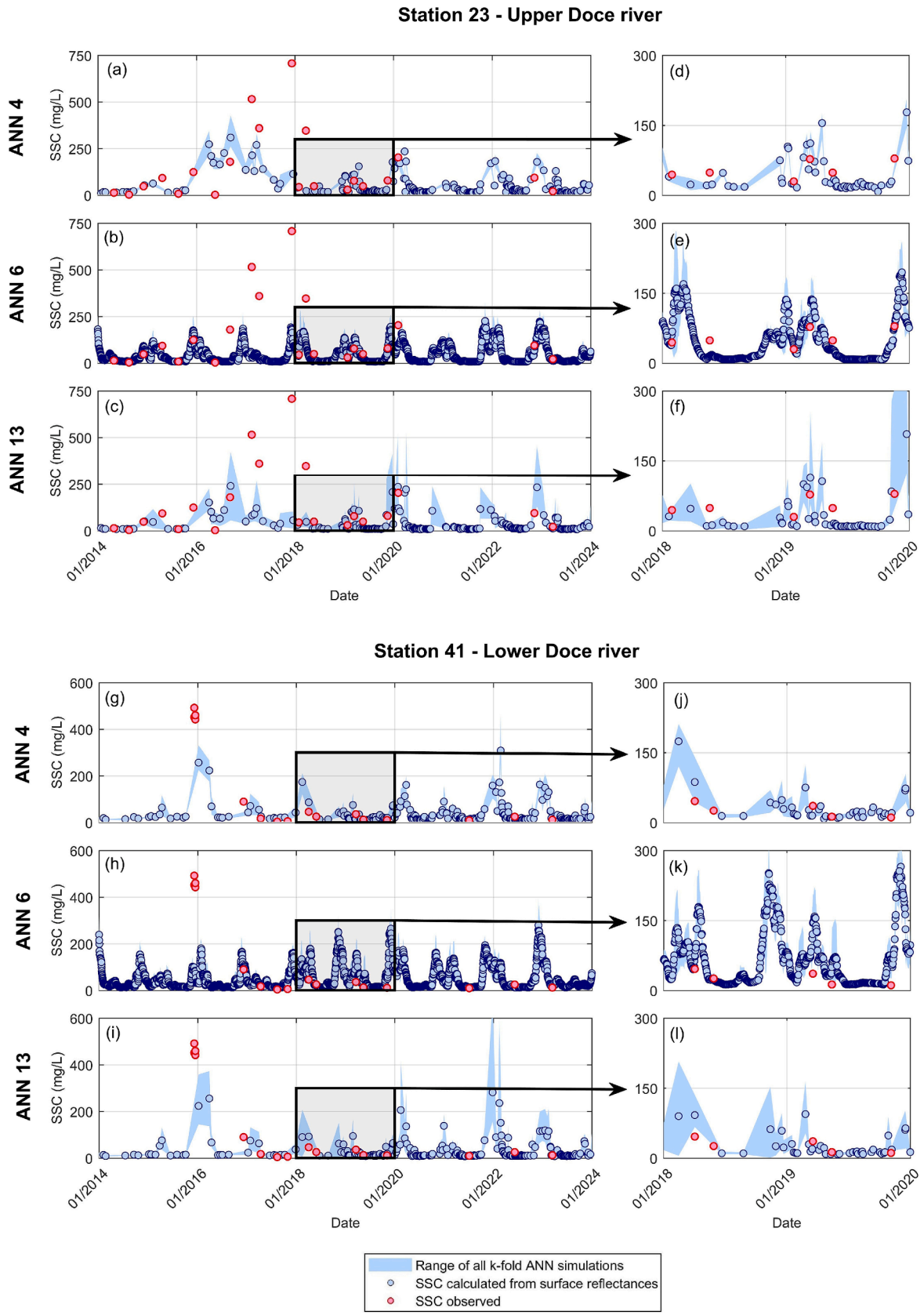


Fig. 8. SSC time series at Station 23 (Upper Doce River) and Station 41 (Lower Doce River) using ANN4, ANN6, and ANN13. Panels (a)–(c) refer to Station 23 and (g)–(i) to Station 41. Blue circles show SSC estimates, red circles are observed SSC, and the blue shaded envelope represents the k-fold simulation range. Zoomed-in views of the highlighted periods are shown in (d)–(f) for Station 23 and (j)–(l) for Station 41. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

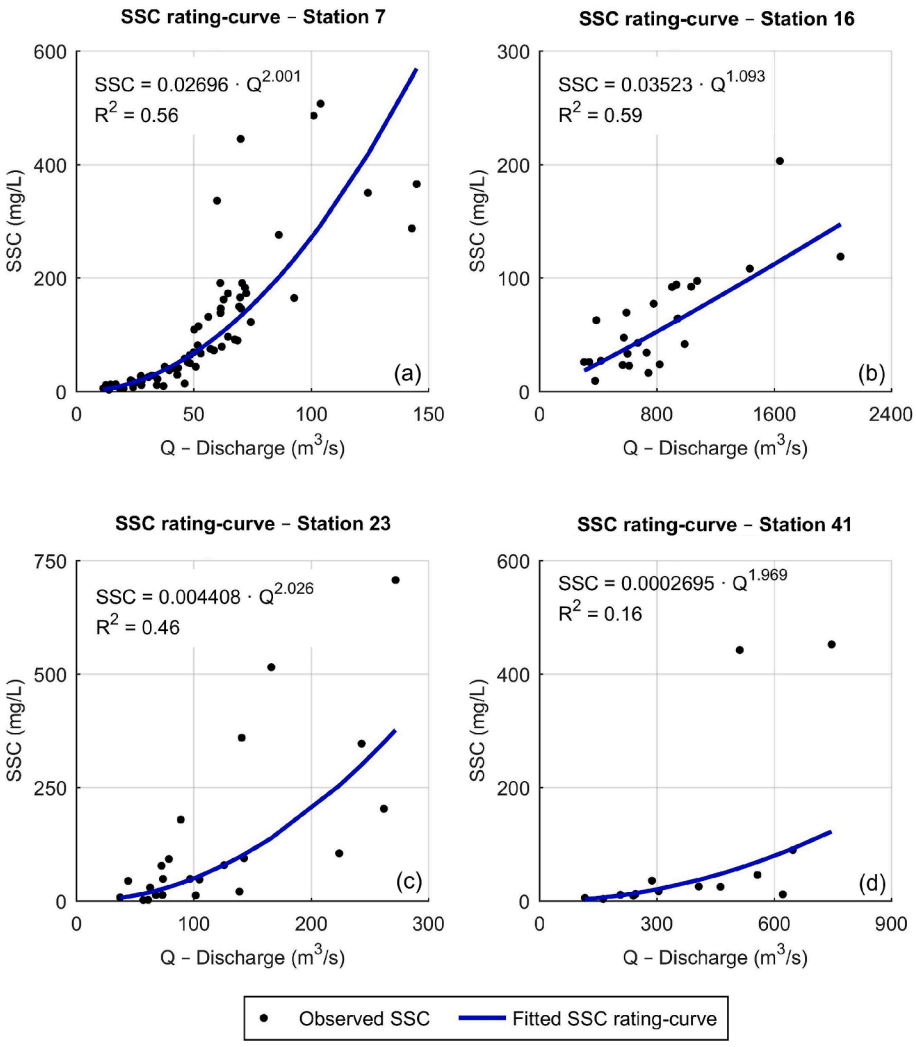


Fig. 9. SSC rating curves fitted at the four gauging stations with concomitant discharge and SSC measurements.

with the behavior of large river systems, where greater channel width, storage capacity, and longer water–sediment travel times tend to moderate and delay seasonal fluctuations in sediment concentration.

4. Discussions

As demonstrated by the results, the three SSC estimation approaches based on remotely sensed variables and ANN models proved effective in enhancing sediment monitoring in regions where in situ data are scarce. Although the two studied river basins contain 41 gauging stations distributed along their main channels, the temporal coverage of SSC measurements at each site is extremely limited. This lack of consistent sampling severely constrains our ability to understand and evaluate sediment dynamics at both spatial and temporal scales, as reported previously in several studies (Campos and Pedrollo, 2021; Cohen et al., 2013; Fagundes et al., 2021, 2020; Pandey et al., 2016; Riveros-Iregui et al., 2018).

In general, all models reproduce the seasonal variability of SSC associated with the rainfall regime. Nevertheless, some high SSC values were not well represented, particularly by the precipitation-based model and, to a lesser extent, by the reflectance-based models, because they were linked to tailings-dam failures. Precipitation-based models cannot anticipate SSC surges driven by non-hydrometeorological causes, whereas reflectance-based models can, but only when cloud-free imagery is available. These approaches are therefore complementary. During the wet season, when SSC is typically highest and optical data gaps are frequent, precipitation-driven models provide regular estimates that are valuable for assessing seasonal SSC dynamics, erosion and sedimentation patterns, and reservoir siltation. Conversely, reflectance-based models enable spatially explicit, pixel-by-pixel assessments and can detect event-driven SSC spikes unrelated to rainfall. Conceptually, the former is anchored in hydrological process drivers (rainfall and runoff, plus sediment transport), while the latter relies on the empirical relationship between water-leaving reflectance and the amount of SSC in the water column.

Taken together, our results show a clear and useful complementarity among the models. Reflectance-based approaches (ANN4 and ANN13) are most sensitive to event-driven SSC surges and enable pixel-level mapping along river corridors, whereas the precipitation-driven model (ANN6) provides stable daily estimates during cloudy periods, capturing seasonal dynamics when optical data are unavailable. A comparison with traditional SSC rating-curves also demonstrated the superior performance of models based on remote-sensing variables, particularly reflectance-based ANNs, while even the precipitation- and soil-driven ANN6 outperformed the rating-curve method at all evaluated stations.

Across sites, ANN13 offers the best overall trade-off between skill and robustness, reconciling optical and hydrometeorological controls. Remaining mismatches are largely associated with persistent cloud cover, sparse local training data, or disturbances not represented in the inputs. From an operational standpoint, the results suggest a hybrid

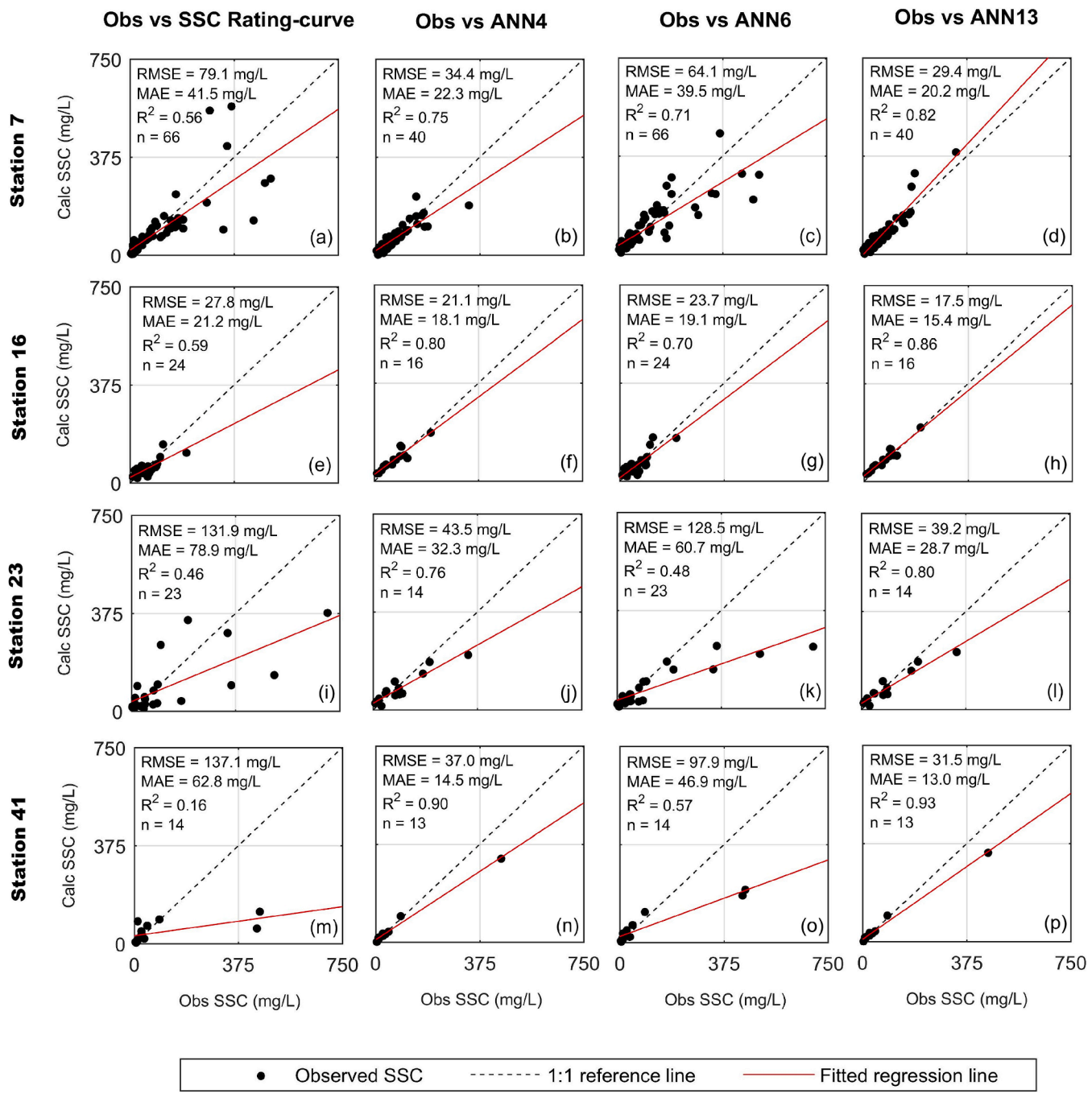


Fig. 10. Comparison between observed SSC and estimated SSC using rating curves and the three ANN models (ANN4, ANN6, ANN13) at four gauging stations.

monitoring workflow: ANN6 can serve as a daily baseline model, reflectance-based models can update estimates opportunistically on cloud-free days, and ANN13 can be used periodically to harmonize predictions by integrating information from both data domains. Implemented within platforms such as Google Earth Engine, this workflow becomes readily deployable for routine monitoring, post-event assessment, and decision support in data-scarce regions. Its performance could be further enhanced by incorporating remotely sensed hydrodynamic information—such as river water surface elevation or discharge estimates—offered by missions like SWOT (Surface Water and Ocean Topography), which currently provides WSE measurements for rivers wider than ~60 m and is expected to deliver discharge products in forthcoming releases (Biancamaria et al., 2016).

Several previous studies have demonstrated that multispectral satellite imagery, particularly from Landsat and Sentinel-2, combined with machine learning methods can provide reliable SSC estimates across a wide range of river systems. Extreme learning machines, feed-forward neural networks, and cascade neural networks have been applied to Landsat imagery to estimate SSC. In these applications, the best-performing model achieved R^2 values above 0.9 and accurately captured both high and low SSC with minimal overfitting (Peterson et al., 2018). Model-averaged neural networks using Landsat 8 have been applied alongside random forest and support vector regression to estimate SSC, with all machine-learning approaches outperforming traditional linear regression methods. In these applications, the best models achieved R^2 values between 0.83 and 0.89 (Larson et al., 2021). In highly turbid environments such as the Yellow River, machine-learning approaches based on Sentinel-2 reflectance have also shown

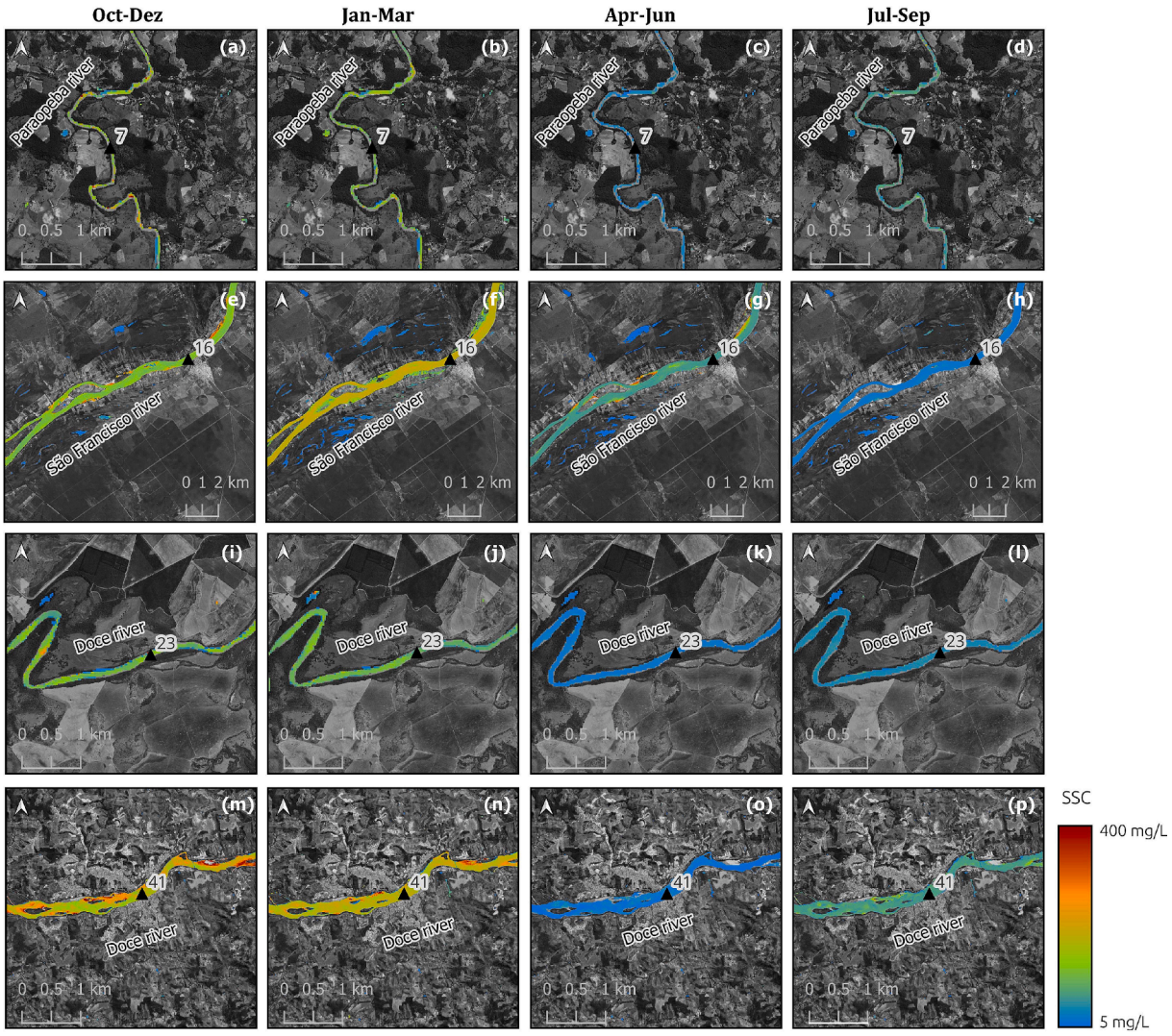


Fig. 11. Quarterly mean SSC maps estimated using ANN13 for four gauging stations: Paraopeba River (station 7), São Francisco River (station 16), and Doce River (stations 23 and 41).

strong performance. Models including random forest and artificial neural networks were tested, and the best-performing configuration achieved R^2 values up to 0.94 when estimating SSC under extremely high sediment loads (Song et al., 2025).

Hybrid approaches combining Landsat and Sentinel-2 imagery have also been applied to estimate SSC in the Lower Brazos River using support vector machines, artificial neural networks, and extreme learning machine models. These multisensor machine-learning configurations produced strong results, with the best models reaching R^2 values between 0.83 and 0.85 (Stull and Ahmari, 2024). More recently, a continental-scale framework integrating 40 years of Landsat surface reflectance with in situ measurements has been developed to estimate SSC across hundreds of rivers in the United States. Using gradient-boosted regression trees and related ensemble learners, the models achieved robust performance at national scale, with R^2 values up to 0.84 (Erten, 2025). Taken together, the performance reported in these studies is consistent with the results obtained here, where multisource artificial neural networks—particularly ANN13—reached testing R^2 values up to 0.84 despite the considerably lower density of in situ SSC observations in our basins.

Importantly, studies conducted in the same region as our work report similar levels of predictive accuracy. Aires et al., (2022) estimated SSC in the Doce River basin using Sentinel-2 and Landsat-8 reflectance combined with linear and regularized regression models, obtaining MAE values of 21–29 mg/L and RMSE values of 28–47 mg/L. These error magnitudes are comparable to those achieved by our reflectance-based models ANN4 and ANN13, which is expected given their shared reliance on surface-reflectance information. However, our framework goes beyond purely optical predictors by incorporating hydrometeorological and soil variables, and by jointly modeling two river basins (the Upper São Francisco and the Doce) to increase the density of in situ observations. This integrated approach enhances the model's ability to generalize across rivers with distinct geomorphological and hydrological characteristics.

When considering studies that integrate hydrometeorological information, very few combine multispectral satellite data with meteorological variables in a unified SSC modeling framework. One recent example is Fang et al., (2025), who proposed a stacking model combining multilayer perceptron, LightGBM, and Elastic Net, integrating Landsat reflectance with precipitation, temperature, wind speed, and surface pressure from ERA5 for long-term USGS stations. Their approach achieved an R^2 of 0.80, illustrating the benefit of incorporating multiple environmental drivers. In contrast, our study employs a simpler set of hydrometeorological predictors, limited primarily to precipitation, yet achieves comparable or slightly higher accuracy (testing R^2 up to 0.84). These results suggest that combining multispectral reflectance

with hydrometeorological variables is a promising direction for improving SSC estimation in data-scarce river systems, especially considering the growing availability of hydrometeorological datasets derived from remote sensing.

Despite these promising directions, several limitations remain. The availability and quality of satellite imagery, particularly during the rainy season, can limit the temporal resolution of reflectance-based models due to cloud cover. Additionally, the performance of the models is inherently influenced by the quality and spatial distribution of in situ SSC data used for training, which remains limited in many regions. The ANN models developed here are basin-specific and may require retraining or adaptation for use in other hydrological or geomorphological contexts.

5. Conclusions

This study compares the potential of using artificial neural networks (ANNs) combined with remote sensing data to estimate SSC in large river basins with limited in situ measurements. Although each individual gauging station provides relatively few SSC samples, their integration into a regional dataset enabled the development of models capable of learning broader sediment dynamics across diverse hydro-geomorphological settings. This integrated approach provides a practical way to address the challenges posed by sparse and temporally limited SSC observations in the region. Models using multispectral reflectance (ANN4), hydrometeorological and soil property information (ANN6), and combined inputs (ANN13) were tested, showing that while reflectance-based models achieved higher accuracy in capturing SSC peaks, precipitation-driven models provided consistent daily estimates aligned with seasonal patterns.

The multisource ANN13 model, integrating both reflectance and environmental predictors, offered a balanced solution capable of capturing seasonal SSC dynamics and, in some cases, short-lived increases in sediment concentration. The ability of these models to generate spatially explicit SSC maps and temporally coherent SSC time series supports improved monitoring and management of sediment dynamics, contributing to water-quality assessments in data-scarce regions.

Overall, the integration of ANNs with remote sensing datasets presents a scalable, low-cost approach for continuous SSC monitoring, enhancing the capacity to understand and manage sediment transport processes in large river systems.

Future studies should explore the transferability of the proposed models to other basins, including diverse data-scarce regions or transboundary rivers. Incorporating additional inputs, such as hydrodynamic variables (e.g., discharge, flow velocity) and land use changes, may enhance model accuracy and responsiveness to anthropogenic pressures. In this context, the forthcoming discharge data products from the SWOT satellite mission are expected to provide valuable high-resolution flow information, which could be integrated into SSC estimation models to improve their representation of sediment transport dynamics.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used ChatGPT (OpenAI) to improve the readability and language of the manuscript. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

CRediT authorship contribution statement

Juliana Andrade Campos: Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Cintia B. Uvo:** Writing – review & editing, Supervision, Methodology, Investigation, Conceptualization. **Thais Fujita:** Writing – review & editing, Validation, Methodology, Formal analysis. **Luz Adriana Cuartas:** Writing – review & editing, Methodology, Conceptualization. **Paulo Rógenes Monteiro Pontes:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

References

Adnan, R.M., Liang, Z., El-Shafie, A., Zounemat-Kermani, M., Kisi, O., 2019. Prediction of suspended sediment load using data-driven models. Water 11, 2060. https://doi.org/10.3390/w11102060.

Aires, U.R.V., da Silva, D.D., Fernandes Filho, E.I., Rodrigues, L.N., Uliana, E.M., Amorim, R.S.S., Ribeiro, C.B. de M., Campos, J.A., 2023. Machine learning-based modeling of surface sediment concentration in Doce river basin. J. Hydrol. 619, 129320. https://doi.org/10.1016/j.jhydrol.2023.129320.

Aires, U.R.V., da Silva, D.D., Fernandes Filho, E.I., Rodrigues, L.N., Uliana, E.M., Amorim, R.S.S., Ribeiro, C.B. de M., Campos, J.A., 2022. Modeling of surface sediment concentration in the Doce River basin using satellite remote sensing. J. Environ. Manage. 323, 116207. https://doi.org/10.1016/j.jenvman.2022.116207.

Al Khoury, I., Boithias, L., Labat, D., 2023. A review of the application of the soil and water assessment tool (SWAT) in karst watersheds. Water 15, 954. https://doi.org/10.3390/w15050954.

Aldin Shojaeezadeh, S., Al-Wardy, M., Reza Nikoo, M., 2024. Suspended sediment load modeling using hydro-climate variables and machine learning. J. Hydrol. 633, 130948. https://doi.org/10.1016/j.jhydrol.2024.130948.

Aloui, S., Mazzoni, A., Elomri, A., Aouissi, J., Boufekane, A., Zghibi, A., 2023. A review of Soil and Water Assessment Tool (SWAT) studies of Mediterranean catchments: applications, feasibility, and future directions. J. Environ. Manage. 326, 116799. https://doi.org/10.1016/j.jenvman.2022.116799.

Alvares, C.A., Stape, J.L., Sentelhas, P.C., de Moraes Gonçalves, J.L., Sparovek, G., 2013. Köppen's climate classification map for Brazil. Meteorol. Zeitschrift 22, 711–728. https://doi.org/10.1127/0941-2948/2013/0507.

Asselman, N.E.M., 2000. Fitting and interpretation of sediment rating curves. J. Hydrol. 234, 228–248. https://doi.org/10.1016/S0022-1694(00)00253-5.

Batjes, N.H., Ribeiro, E., van Oostrum, A., 2020. Standardised soil profile data to support global mapping and modelling (WoSIS snapshot 2019). Earth Syst. Sci. Data 12, 299–320. https://doi.org/10.5194/essd-12-299-2020.

Bettencourt, P., Fulgêncio, C., Grade, M., Alcobia, S., Monteiro, J.P., Oliveira, R., Leitão, J.C., Leitão, P.C., Fernandes, P.A., de Sousa, S., Brites, S., Fernandes, J., Simões, J., Scarton, M., Santiago, E., Aguiar, R., Giffoni, M., Melo, F., Paes, A.C., 2016. Plano de recursos hídricos da bacia hidrográfica do rio São Francisco. Rev. Recur. Hídricos 37, 73–80. https://doi.org/10.5894/rh37n1-cti3.

Biancamaria, S., Lettenmaier, D.P., Pavelsky, T.M., 2016. The SWOT mission and its capabilities for land hydrology. Remote Sens. Water Resour. 117–147.

Buyukyildiz, M., Kumcu, S.Y., 2017. An estimation of the suspended sediment load using adaptive network based fuzzy inference system, support vector machine and artificial neural network models. Water Resour. Manag. 31, 1343–1359. https://doi.org/10.1007/s11269-017-1581-1.

Campos, J.A., Pedrollo, O.C., 2023. Modelling and assessing how small hydropower facilities affect sediment transport by using fuzzy inference systems. J. Hydrol. 620, 129374. https://doi.org/10.1016/j.jhydrol.2023.129374.

Campos, J.A., Pedrollo, O.C., 2021. A regional ANN-based model to estimate suspended sediment concentrations in ungauged heterogeneous basins. Hydrol. Sci. J. 66, 1222–1232. https://doi.org/10.1080/02626667.2021.1918695.

Cao, L., Liu, S., Wang, S., Cheng, Q., Fryar, A.E., Zhang, L., Zhang, Z., Yue, F., Peng, T., 2021. Factors controlling discharge-suspended sediment hysteresis in karst basins, southwest China: implications for sediment management. J. Hydrol. 594, 125792. https://doi.org/10.1016/j.jhydrol.2020.125792.

Choubin, B., Darabi, H., Rahmati, O., Sajedi-Hosseini, F., Kløve, B., 2018. River suspended sediment modelling using the CART model: a comparative study of

machine learning techniques. Sci. Total Environ. 615, 272–281. https://doi.org/10.1016/j.scitotenv.2017.09.293.
Cohen, S., Kettner, A.J., Syvitski, J.P.M., Fekete, B.M., 2013. WBMsed, a distributed global-scale riverine sediment flux model: model description and validation. Comput. Geosci. 53, 80–93. https://doi.org/10.1016/j.cageo.2011.08.011.
Costa, F., Dela, P., Buarque, D.C., Brêda, J.P.L.F., Föeger, L.B., 2022. Impact of climate change on the flow of the Doce River basin. RBRH 27. https://doi.org/10.1590/2318-0331.272220220069.
Dethier, E.N., Renshaw, C.E., Magilligan, F.J., 2022. Rapid changes to global river suspended sediment flux by humans. Science (80-.) 376, 1447–1452. https://doi.org/10.1126/science.abn7980.
Domingues, L.M., 2011. Gestão integrada de recursos hídricos-o caso da bacia hidrográfica do Rio Doce.
Duan, M., Qiu, Z., Li, R., Li, K., Yu, S., Liu, D., 2024. Monitoring suspended sediment transport in the Lower Yellow River using landsat observations. Remote Sens. 16, 229. https://doi.org/10.3390/rs16020229.
Erten, G., 2025. A machine learning framework for estimating suspended sediment continental United States using 40 years of Landsat observations. Sci. Total Environ. 1006, 180941. https://doi.org/10.1016/j.scitotenv.2025.180941.
Fagundes, H. de O., Paiva, R.C.D., Brêda, J.P.L.F., Fassoni-Andrade, A.C., Borrelli, P., Fan, F.M., 2023. An assessment of South American sediment fluxes under climate changes. Sci. Total Environ. 879, 163056. https://doi.org/10.1016/j.scitotenv.2023.163056.
Fagundes, H.O., Fan, F.M., Paiva, R.C.D., Siqueira, V.A., Buarque, D.C., Kornowski, L.W., Laipelt, L., Collischonn, W., 2021. Sediment flows in South America supported by daily hydrologic-hydrodynamic modeling. Water Resour. Res. 57. https://doi.org/10.1029/2020WR027884.
Fagundes, H.O., Paiva, R.C.D., Fan, F.M., Buarque, D.C., Fassoni-Andrade, A.C., 2020. Sediment modeling of a large-scale basin supported by remote sensing and in-situ observations. Catena 190, 104535. https://doi.org/10.1016/j.catena.2020.104535.
Fang, X., Zhang, J., Yu, X., Zhang, S., Kong, D., Wang, X., Ali, S., Ullah, H., Xu, N., 2025. Accurate estimation of suspended sediment concentration integrated remote sensing information and a novel stacking machine learning model. Stoch. Environ. Res. Risk Assess. 39, 1517–1535. https://doi.org/10.1007/s00477-025-02930-4.
Fassoni-Andrade, A.C., de Paiva, R.C.D., 2019. Mapping spatial-temporal sediment dynamics of river-floodplains in the Amazon. Remote Sens. Environ. 221, 94–107. https://doi.org/10.1016/j.rse.2018.10.038.
Fassoni-Andrade, A.C., Fleischmann, A.S., Papa, F., de Paiva, R.C.D., Wongchuig, S., Melack, J.M., Moreira, A.A., Paris, A., Ruhoff, A., Barbosa, C., Maciel, D.A., Novo, E., Durand, F., Frappart, F., Aires, F., Abrahão, G.M., Ferreira-Ferreira, J., Espinoza, J. C., Laipelt, L., Costa, M.H., Espinoza-Villar, R., Calmant, S., Pellet, V., 2021. Amazon hydrology from space: scientific advances and future challenges. Rev. Geophys. 59. https://doi.org/10.1029/2020RG000728.
Foga, S., Scaramuzza, P.L., Guo, S., Zhu, Z., Dilley Jr., R.D., Beckmann, T., et al., 2017. Cloud detection algorithm comparison and validation for operational Landsat data products. Remote Sens Environ 194, 379–390. https://doi.org/10.1016/j.rse.2017.03.026.
Funk, C., Peterson, P., Landsfeld, M., Pedreros, D., Verdin, J., Shukla, S., Husak, G., Rowland, J., Harrison, L., Hoell, A., Michaelsen, J., 2015. The climate hazards infrared precipitation with stations—a new environmental record for monitoring extremes. Sci. Data 2, 150066. https://doi.org/10.1038/sdata.2015.66.
Gebremicael, T.G., Mohamed, Y.A., van der Zaag, P., Berhe, A.G., Haile, G.G., Hagos, E. Y., Hagos, M.K., 2017. Comparison and validation of eight satellite rainfall products over the rugged topography of Tekeze-Atbara Basin at different spatial and temporal scales. Hydrol. Earth Syst. Sci. Discuss. 1–31. https://doi.org/10.5194/hess-2017-504.
Hecht-Nielsen, R., 1990. Neurocomputing. Addison-Wesley Publishing Company, Massachusetts.
Heng, S., Suetsugi, T., 2013. Using artificial neural network to estimate sediment load in ungauged catchments of the Tonle Sap River Basin. Cambodia. J. Water Resour. Prot. 05, 111–123. https://doi.org/10.4236/jwarp.2013.52013.
Javed, A., Hamshaw, S.D., Lee, B.S., Rizzo, D.M., 2021. Multivariate event time series analysis using hydrological and suspended sediment data. J. Hydrol. 593, 125802. https://doi.org/10.1016/j.jhydrol.2020.125802.
Jing, T., Zeng, Y., Fang, N., Dai, W., Shi, Z., 2025. A review of suspended sediment hysteresis. Water Resour. Res. 61. https://doi.org/10.1029/2024WR037216.
Kabir, S.M.I., Ahmari, H., 2020. Evaluating the effect of sediment color on water radiance and suspended sediment concentration using digital imagery. J. Hydrol. 589, 125189. https://doi.org/10.1016/j.jhydrol.2020.125189.
Kay, S., Hedley, J., Lavender, S., 2009. Sun glint correction of high and low spatial resolution images of aquatic scenes: a review of methods for visible and near-infrared wavelengths. Remote Sens. 1, 697–730. https://doi.org/10.3390/rs1040697.
Kuhn, C., de Matos Valerio, A., Ward, N., Loken, L., Sawakuchi, H.O., Kampel, M., Richey, J., Stadler, P., Crawford, J., Striegl, R., Vermote, E., Pahlevan, N., Butman, D., 2019. Performance of Landsat-8 and Sentinel-2 surface reflectance products for river remote sensing retrievals of chlorophyll-a and turbidity. Remote Sens. Environ. 224, 104–118. https://doi.org/10.1016/j.rse.2019.01.023.
Lamane, H., Mouhir, L., Moussadek, R., Baghdad, B., Kisi, O., El Bilali, A., 2025. Interpreting machine learning models based on SHAP values in predicting suspended sediment concentration. Int. J. Sedim. Res. 40, 91–107. https://doi.org/10.1016/j.ijsrc.2024.10.002.
Larson, M.D., Simic Milas, A., Vincent, R.K., Evans, J.E., 2021. Landsat 8 monitoring of multi-depth suspended sediment concentrations in Lake Erie's Maumee River using machine learning. Int. J. Remote Sens. 42, 4064–4086. https://doi.org/10.1080/01431161.2021.1890268.
Lehner, B., Grill, G., 2013. Global river hydrography and network routing: baseline data and new approaches to study the world's large river systems. Hydrol. Process. 27, 2171–2186. https://doi.org/10.1002/hyp.9740.
Lloyd, C.E.M., Freer, J.E., Johnes, P.J., Collins, A.L., 2016. Using hysteresis analysis of high-resolution water quality monitoring data, including uncertainty, to infer controls on nutrient and sediment transfer in catchments. Sci. Total Environ. 543, 388–404. https://doi.org/10.1016/j.scitotenv.2015.11.028.
Lohani, A.K., Goel, N.K., Bhatia, K.K.S., 2007. Deriving stage-discharge-sediment concentration relationships using fuzzy logic. Hydrol. Sci. J. https://doi.org/10.1623/hysj.52.4.793.
Medeiros, E.S. de, Oliveira, W. dos S., Silva, P.V. da, Souza, A. de, Bicalho, C.C., Mauad, M., Souza, S.A. de, 2024. Mapping rainfall variability in the são francisco river basin: insights for water resource management. Rev. Bras. Meteorol. 39. https://doi.org/10.1590/0102-77863910050.
Mohamadi, M.A., Kavian, A., 2015. Effects of rainfall patterns on runoff and soil erosion in field plots. Int. Soil Water Conserv. Res. 3, 273–281. https://doi.org/10.1016/j.iswcr.2015.10.001.
Mollaee, Z., Zahiri, J., Jalili, S., Ansari, M.R., Taghizadeh, A., 2018. Estimating suspended sediment concentration using remote sensing and artificial neural network (Case Study: Karun River). J. Water Soil Sci. 22, 249–259. https://doi.org/10.29252/jstnar.22.2.249.
Muller-Wilm, U., 2012. Sentinel-2 MSI—level 2A products algorithm theoretical basis document, ESA, Paris, France. Tech. Rep. S2PAD-ATBD-0001.
Nguyen, S.Q., Nguyen, L.C., Ngo-Duc, T., Ouillon, S., 2024. Applying a machine learning-based method for the prediction of suspended sediment concentration in the Red river basin. Model. Earth Syst. Environ. 10, 2675–2692. https://doi.org/10.1007/s40808-023-01915-y.
Nourani, V., Kalantari, O., Baghanam, A.H., 2012. Two semidistributed ANN-based models for estimation of suspended sediment load. J. Hydrol. Eng. 17, 1368–1380. https://doi.org/10.1061/(ASCE)HE.1943-5584.0000587.
Pandey, A., Himanshu, S.K., Mishra, S.K., Singh, V.P., 2016. Physically based soil erosion and sediment yield models revisited. Catena 147, 595–620. https://doi.org/10.1016/j.catena.2016.08.002.
Pereira, F.J.S., Costa, C.A.G., Foerster, S., Brosinsky, A., de Araújo, J.C., 2019. Estimation of suspended sediment concentration in an intermittent river using multi-temporal high-resolution satellite imagery. Int. J. Appl. Earth Obs. Geoinf. 79, 153–161. https://doi.org/10.1016/j.jag.2019.02.009.
Peterson, K., Sagan, V., Sidike, P., Cox, A., Martinez, M., 2018. Suspended sediment concentration estimation from Landsat imagery along the Lower Missouri and Middle Mississippi rivers using an extreme learning machine. Remote Sens. 10, 1503. https://doi.org/10.3390/rs10101503.
Rahman, S.A., Chakrabarty, D., 2020. Sediment transport modelling in an alluvial river with artificial neural network. J. Hydrol. 588, 125056. https://doi.org/10.1016/j.jhydrol.2020.125056.
Riveros-Iregui, D.A., Covino, T.P., González-Pinzón, R., 2018. The importance of and need for rapid hydrologic assessments in Latin America. Hydrol. Process. 32, 2441–2451. https://doi.org/10.1002/hyp.13163.
Rumelhart, D.E., Hinton, G.E., Williams, R.J., 1986. Learning representations by back-propagating errors. Nature 323, 533–536. https://doi.org/10.1038/323533a0.
Salih, S.Q., Sharafati, A., Khosravi, K., Faris, H., Kisi, O., Tao, H., Ali, M., Yaseen, Z.M., 2020. River suspended sediment load prediction based on river discharge information: application of newly developed data mining models. Hydrol. Sci. J. 65, 624–637. https://doi.org/10.1080/02626667.2019.1703186.
Sari, V., Castro, N.M. dos R., Pedrollo, O.C., 2017. Estimate of suspended sediment concentration from monitored data of turbidity and water level using artificial neural networks. Water Resour. Manag. 31, 4909–4923. https://doi.org/10.1007/s11269-017-1785-4.
Song, G., Jiang, Y., Lei, X., Zhai, S., 2025. Measurement of suspended sediment concentration at the outlet of the Yellow River Canyon: using Sentinel-2 images and machine learning. Remote Sens. 17, 2424. https://doi.org/10.3390/rs17142424.
Stull, T., Ahmari, H., 2024. Estimation of suspended sediment concentration along the Lower Brazos River using satellite imagery and machine learning. Remote Sens. 16, 1727. https://doi.org/10.3390/rs16101727.
Vercruysse, K., Grabowski, R.C., Rickson, R.J., 2017. Suspended sediment transport dynamics in rivers: multi-scale drivers of temporal variation. Earth-Sci. Rev. 166, 38–52. https://doi.org/10.1016/j.earscirev.2016.12.016.
Vermote, E., Roger, J.-C., Franch, B., Skakun, S., 2018. LaSRC (Land Surface Reflectance Code): Overview, application and validation using MODIS, VIIRS, LANDSAT and Sentinel 2 data's. In: IGARSS 2018-2018 IEEE International Geoscience and Remote Sensing Symposium. IEEE, pp. 8173–8176.
Yu, B., Shi, Z., Zhang, Y., 2023. Linking hydrological and landscape characteristics to suspended sediment-discharge hysteresis in Wudinghe River Basin on the Loess Plateau, China. CATENA 228, 107169. https://doi.org/10.1016/j.catena.2023.107169.
Zhang, W., Wei, X., Jinhai, Z., Yuliang, Z., Zhang, Y., 2012. Estimating suspended sediment loads in the Pearl River Delta region using sediment rating curves. Cont. Shelf Res. 38, 35–46. https://doi.org/10.1016/j.csr.2012.02.017.